

GamutX

Workspace Setup

GIT

Workspace Setup

- Python Installation
 - Download Python 3.7.9 / 3.10.11 / 3.13.3
 - Run the Installer
 - Verify the Installation
- Git Installation
- IDE Installation (VS Code)
 - Configuring Visual Studio Code
 - Installing Initial Extensions
 - Opening a Project/Folder in VS Code

Download Python 3.9.10 for **Window**

Download Python

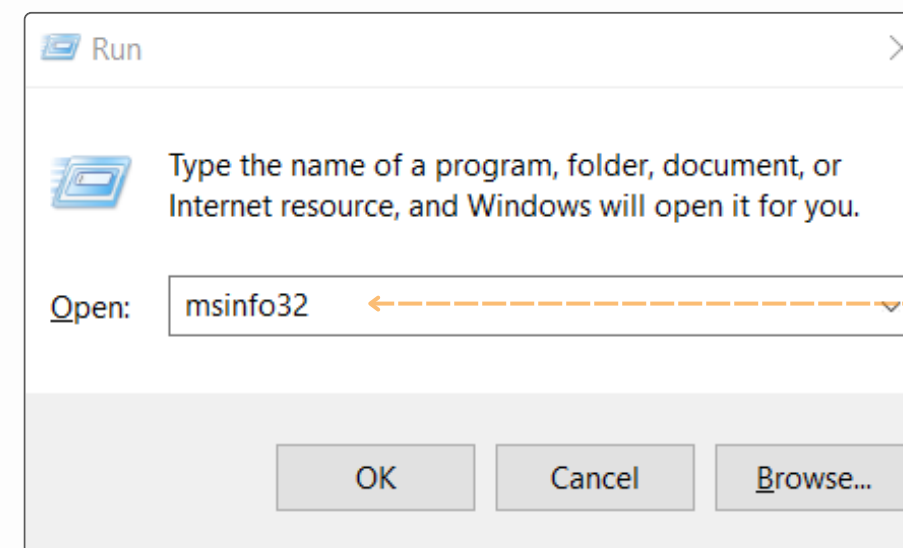
Files			
Version	Operating System	Description	MD5 Sum
Gzipped source tarball	Source release		bcd9f22cf531efc6f06ca6b9b2919b0
XZ compressed source tarball	Source release		389d3ed26b4d97c741d9e5423da1f
macOS 64-bit installer	macOS	for OS X 10.9 and later	4b544fc0ac8c3cfffdb67dede23ddb7
Windows help file	Windows		1094c8d9438ad1adc263ca57ceb3b
Windows x86 embeddable zip file	Windows		97c6558d479dc53bf448580b66ad7
Windows x86 executable installer	Windows		1e6d31c98c68c723541f0821b3c15c
Windows x86 web-based installer	Windows		22f68f09e533c4940fc006e035f08aa
Windows x86-64 embeddable zip file	Windows	for AMD64/EM64T/x64	60f77740b30030b22699dbd14883a
Windows x86-64 executable installer	Windows	for AMD64/EM64T/x64	7083fed513c3c9a4ea655211df9ade
Windows x86-64 web-based installer	Windows	for AMD64/EM64T/x64	da0b17ae84d6579f8df3eb24927fd8

- Search "**Python 3.7.9 Download**" in your browser.
- Click on the **first link**.
- Download the correct installer based on your **System Type**
 - **64-bit** → Windows x86-64 executable installer
 - **32-bit** → Windows x86 executable installer



How to Check Your System Type ?

Press Win + R



Type msinfo32

◆ System Type:

System Model	HP Laptop 15s-du2xxx
System Type	x64-based PC
System SKU	2C467BA#ACI

● **x64-based PC** → 64-bit

● **x86-based PC** → 32-bit

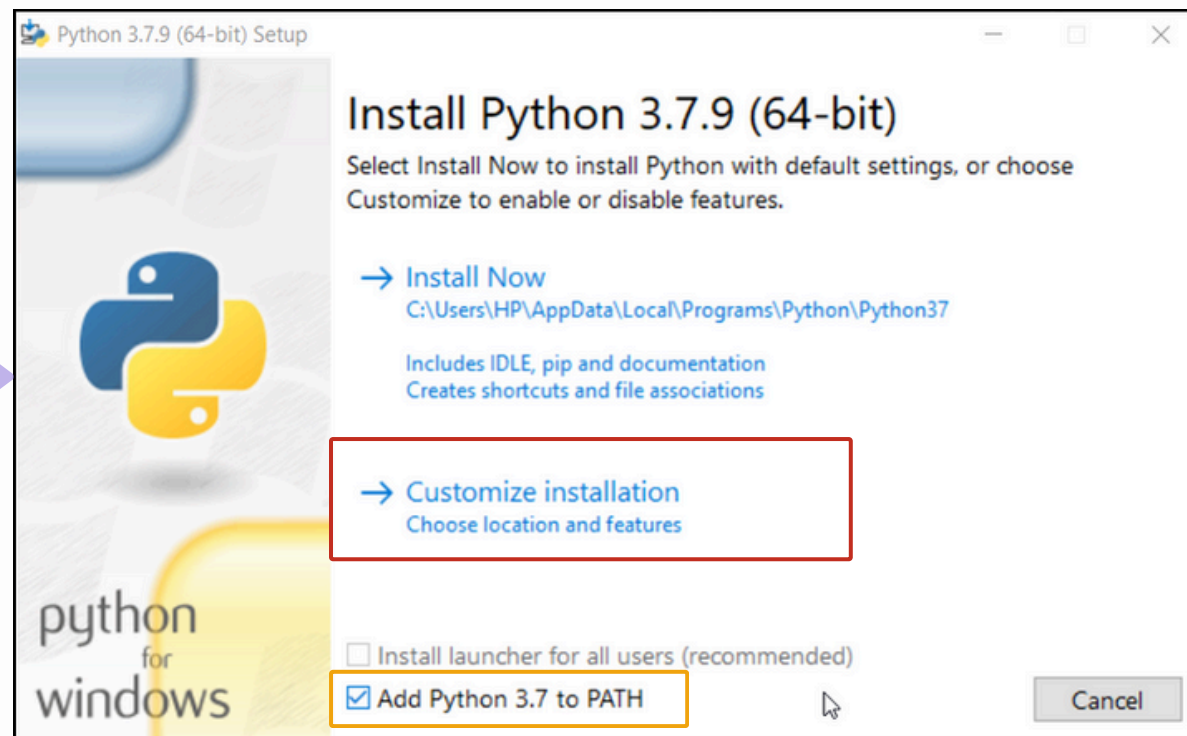
Run the Installer

Run the Installer



Double-click to open

Enable Essential Options

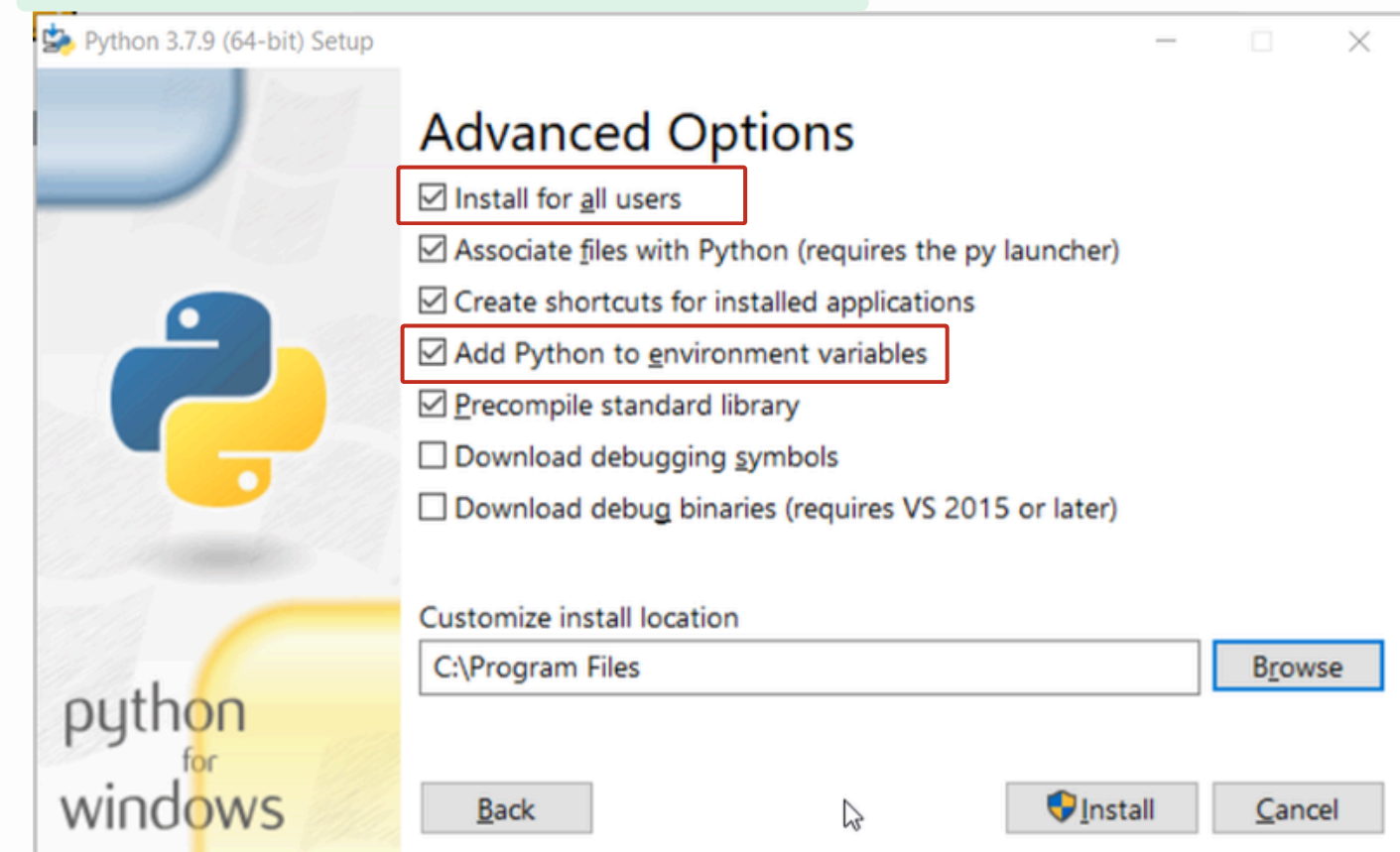


- ✓ Check "**Add Python to PATH**" at the bottom
- ✓ Click **Customize Installation**

Select Optional Features

- ✓ Ensure all **checkboxes** are selected in Optional Feature
- ✓ Click **Next** to continue

Choose Installation Location

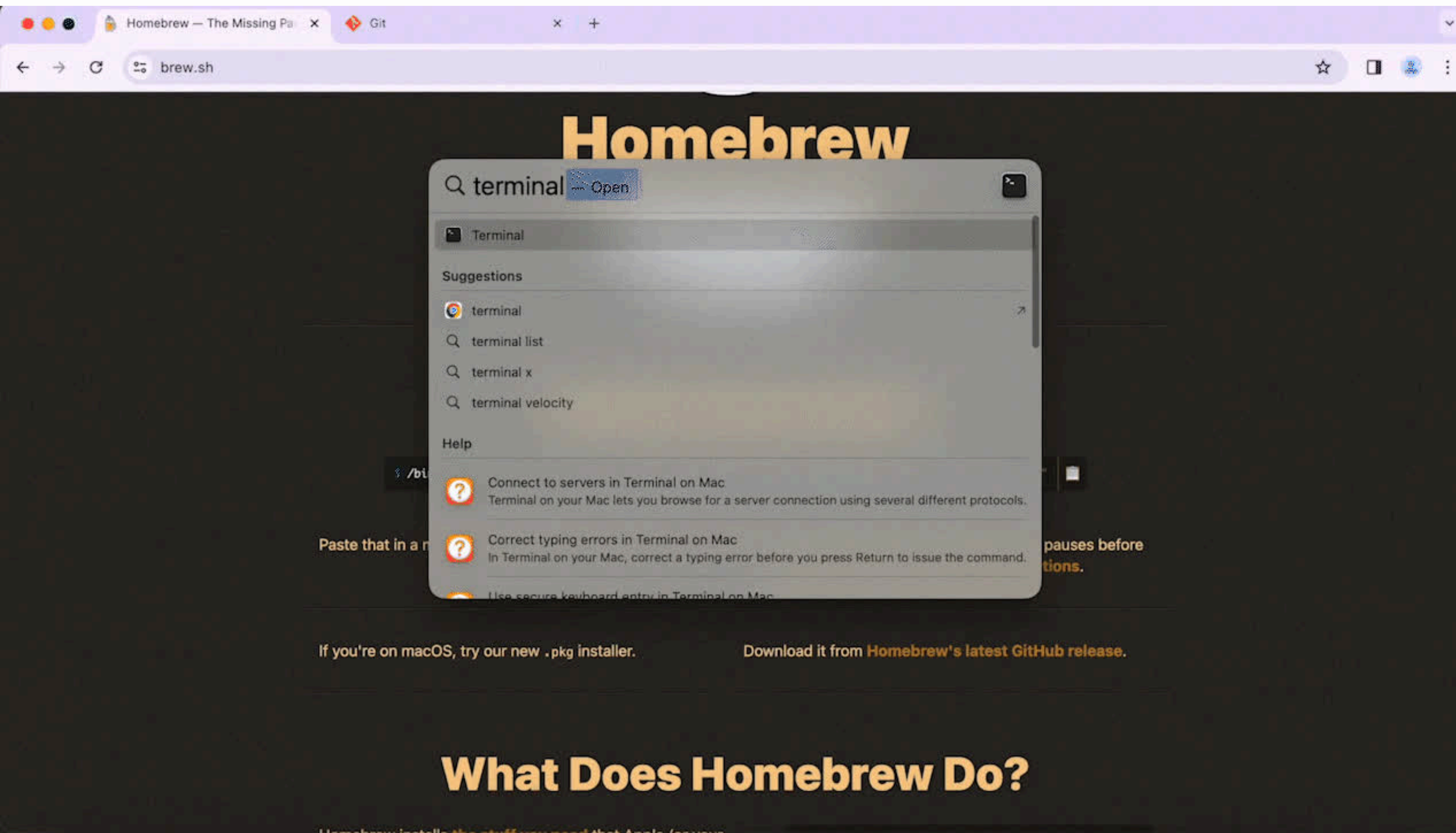


- ! Set path to **C:\Program Files**
- ✓ Check "**Install for all users**"
- ✓ Check "**Add Python to environment variables**"

Download Python 3.7.9 for Mac OS

Installing & Verify Homebrew

- Search **Homebrew** on google



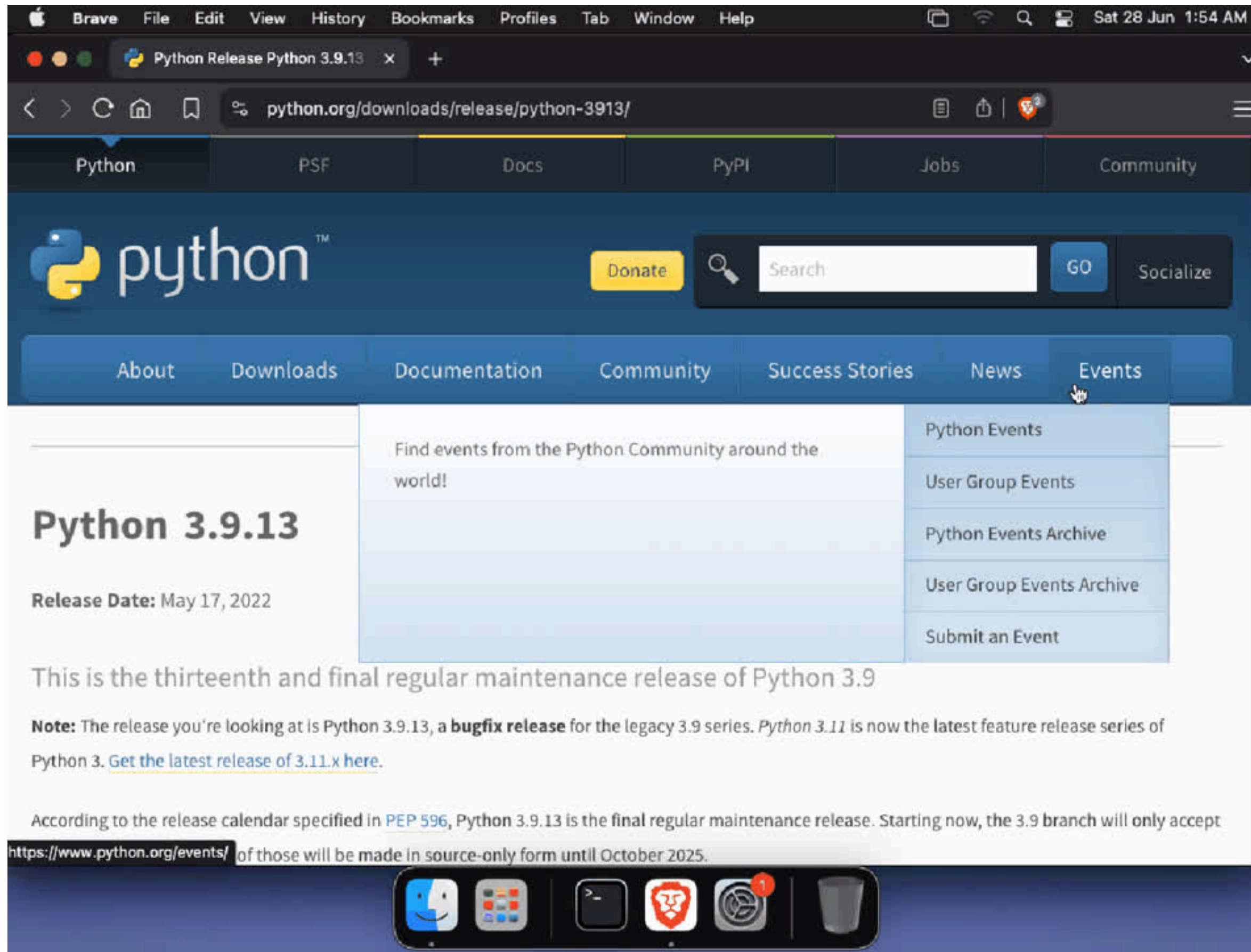
Steps

- Go to brew.sh
- Copy the **install** command
- **Paste** it in **Terminal**
- Enter your **system password**

Run:

- **homebrew --version**
- **to verify installation**

Install Python



Steps

- Search **Python 3.9.13**
- Navigate to **Downloads**
 - **macOS (or your OS)**
- **Download Python installer**
- **Double Click on installer**

After install,

- run: **python3 --version**

Verify PIP

- run : **pip3 --version**

Install GIT

GamutX

Steps

- Go to git-scm.com
- Copy Command
 - **brew install git**
- Open Terminal: paste Command
- Wait for installation to complete
- Verify with:
 - **git --version**

```
pratham — ruby -W1 --disable=gems,rubyopt /opt/homebrew/Library/Ho...

Troubleshooting:
  brew config
  brew doctor
  brew install --verbose --debug FORMULA|CASK

Contributing:
  brew create URL [--no-fetch]
  brew edit [FORMULA|CASK...]

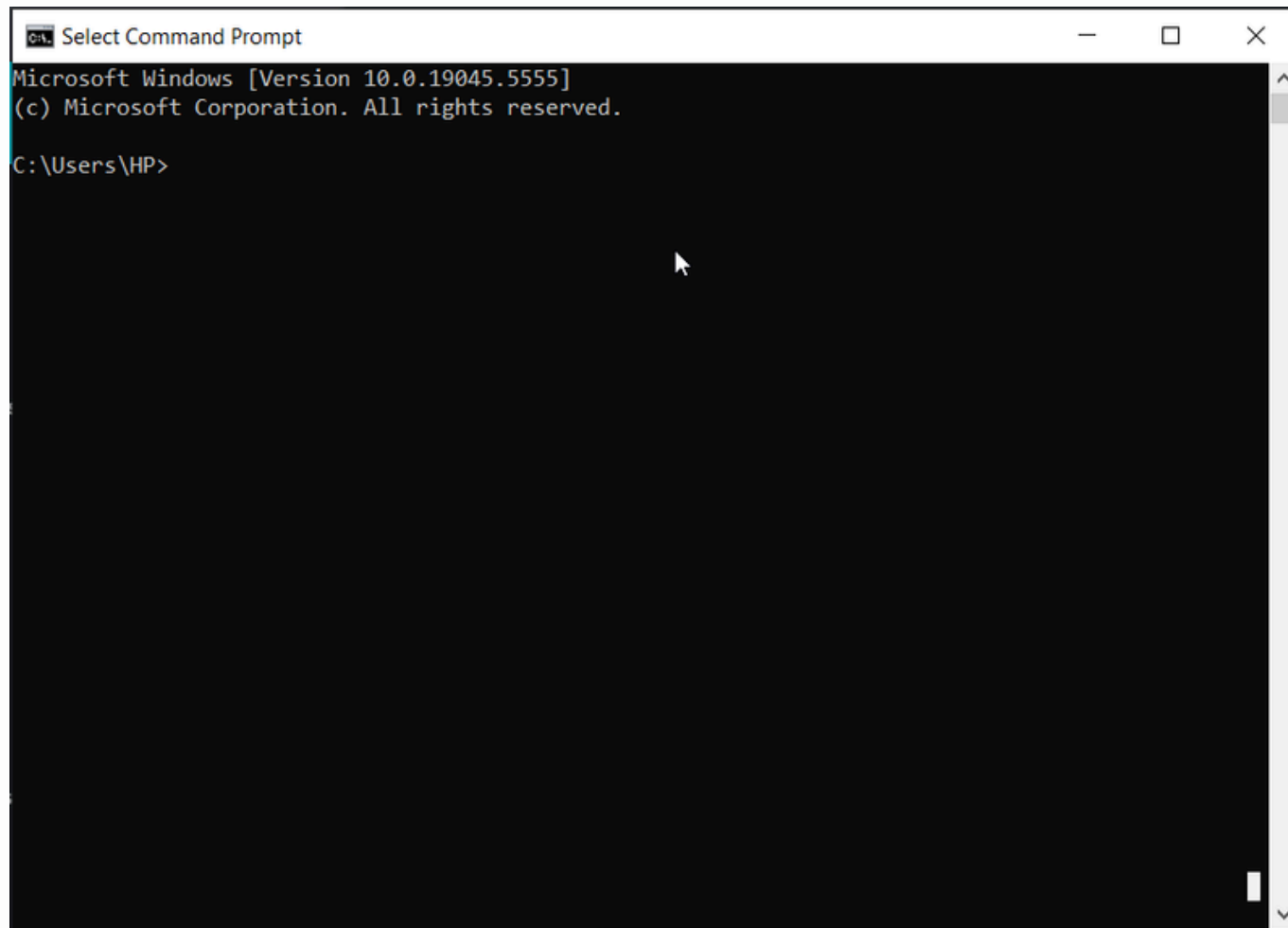
Further help:
  brew commands
  brew help [COMMAND]
  man brew
  https://docs.brew.sh

[pratham@Prathams-MacBook-Air ~ % brew install git
=> Downloading https://ghcr.io/v2/homebrew/core/git/manifests/2.47.1
##### 100.0%
=> Fetching dependencies for git: libunistring, gettext and pcre2
=> Downloading https://ghcr.io/v2/homebrew/core/libunistring/manifests/1.3
##### 100.0%
=> Fetching libunistring
=> Downloading https://ghcr.io/v2/homebrew/core/libunistring/blobs/sha256:38e44
##=##
```

Understanding CMD & PowerShell



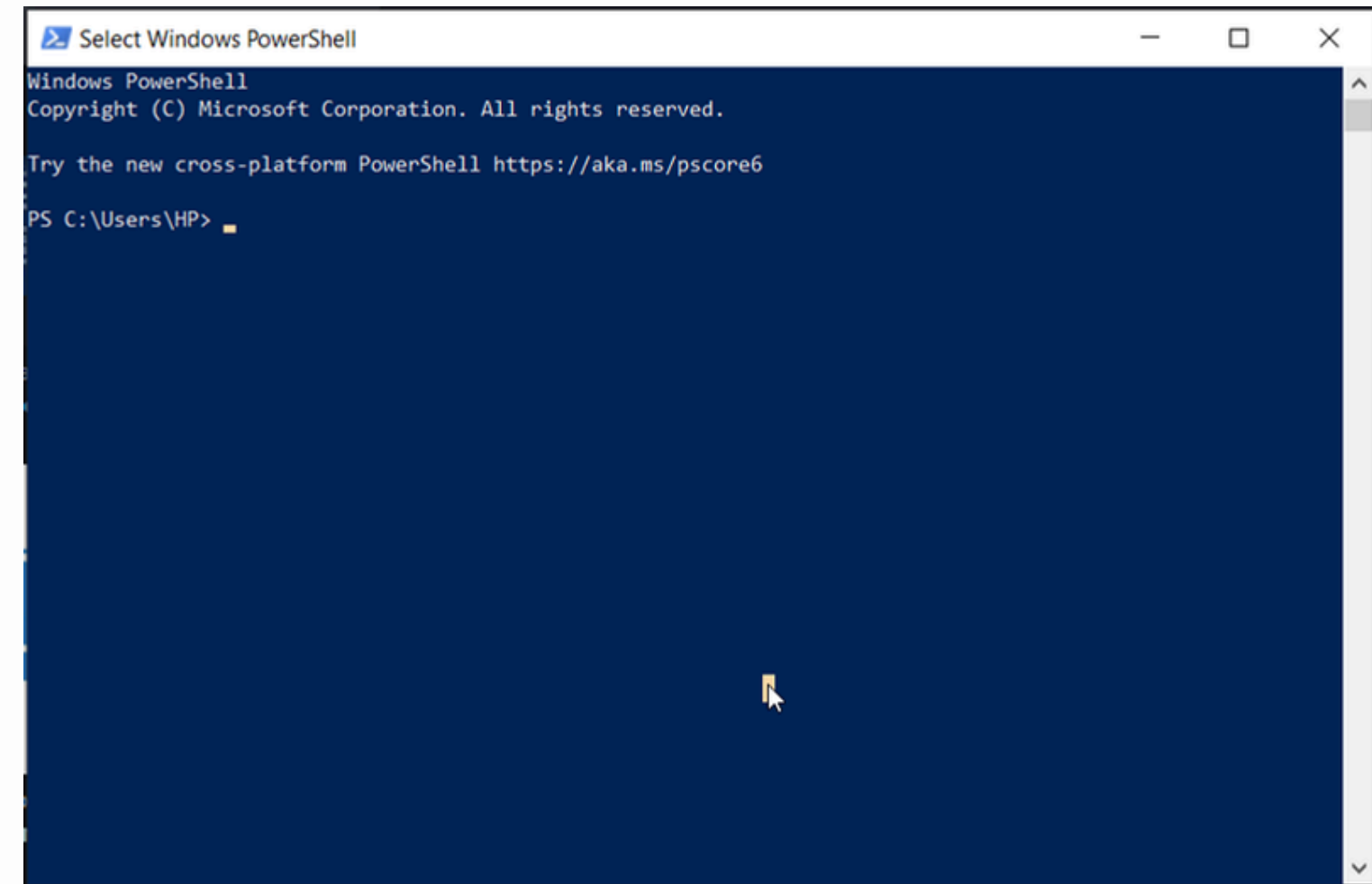
CMD (Command Prompt)



- Basic **command-line interface** for Windows.
- Used for **simple commands** like file **operations & navigation**.
- **Default** shell in **older Windows versions**.



PowerShell

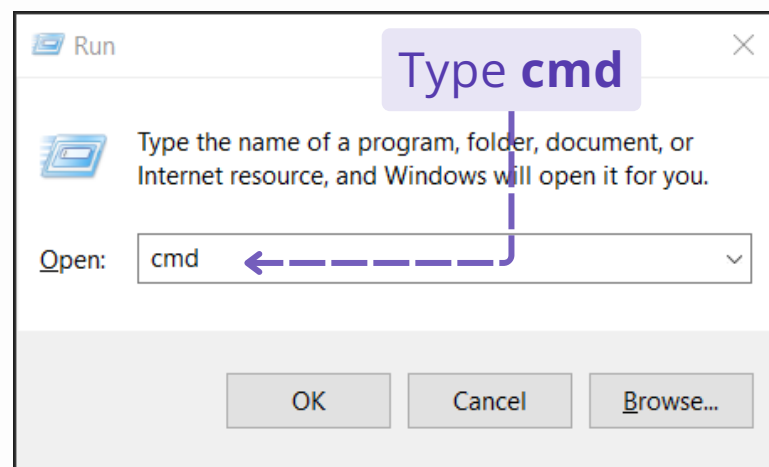


- More **advanced than CMD** – supports **scripting & automation**.
- Uses for **complex tasks**.
- **Built-in support** for system **management & automation**.

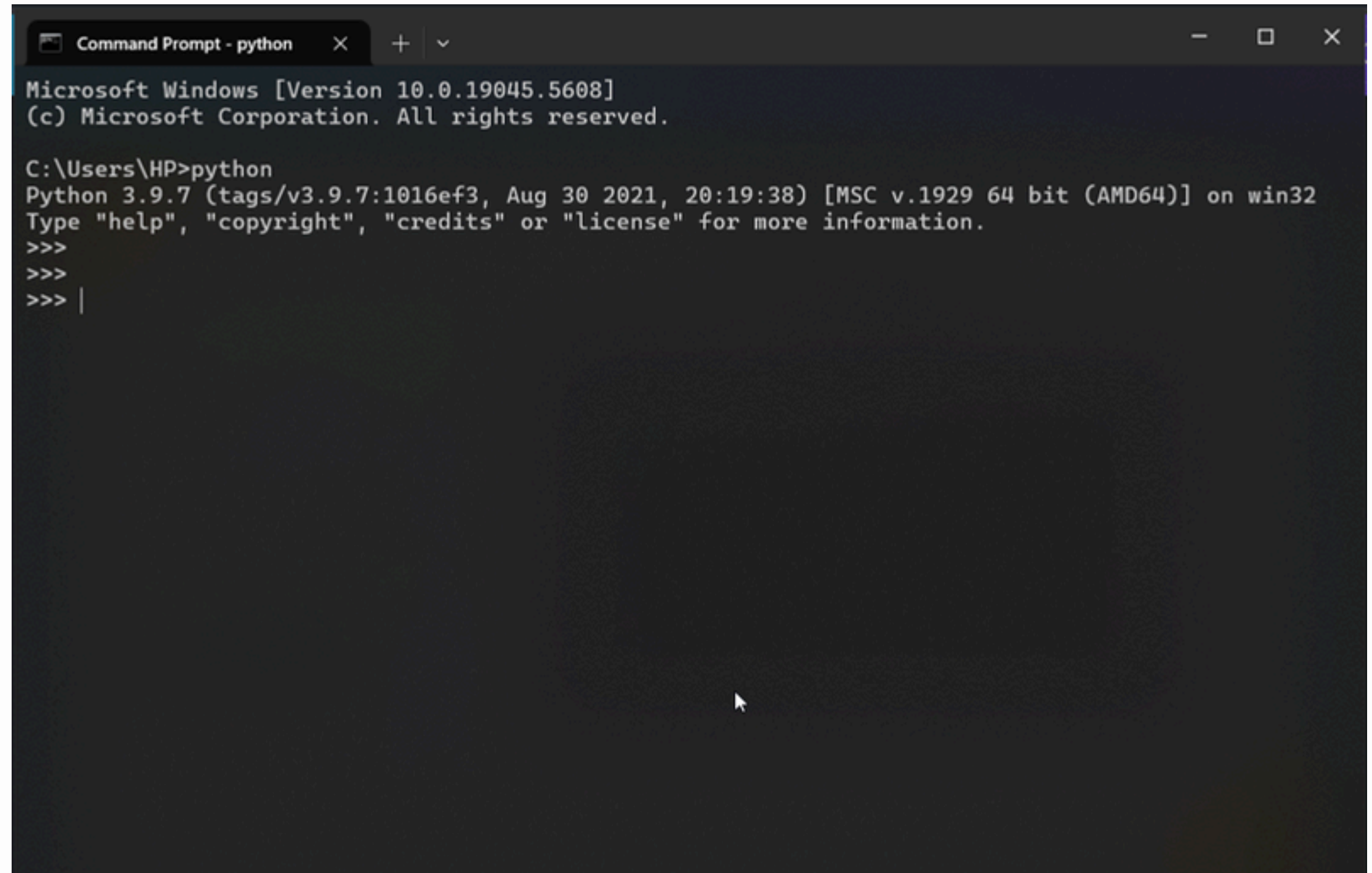
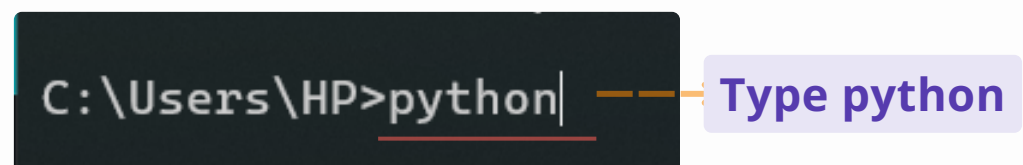
Verify the Installation

Check if Python is Installed

👉 Press Win + R.



On Command Prompt



```
Command Prompt - python
Microsoft Windows [Version 10.0.19045.5608]
(c) Microsoft Corporation. All rights reserved.

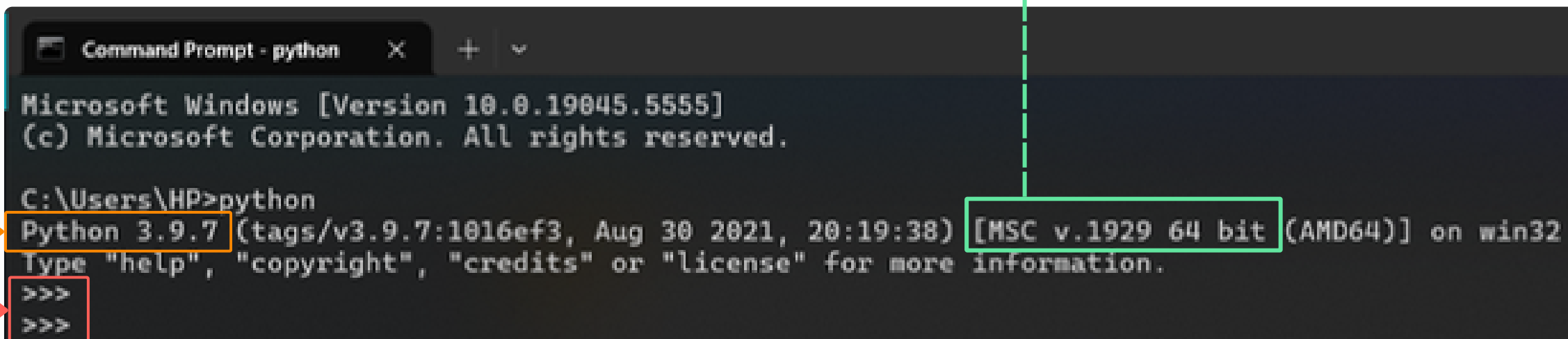
C:\Users\HP>python
Python 3.9.7 (tags/v3.9.7:1016ef3, Aug 30 2021, 20:19:38) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>>
>>>
>>> |
```

Understand the Output

☞ If installed correctly, you will see the **Python interpreter**.

Indicates installed Python version

System architecture



```
Command Prompt - python
Microsoft Windows [Version 10.0.19045.5555]
(c) Microsoft Corporation. All rights reserved.

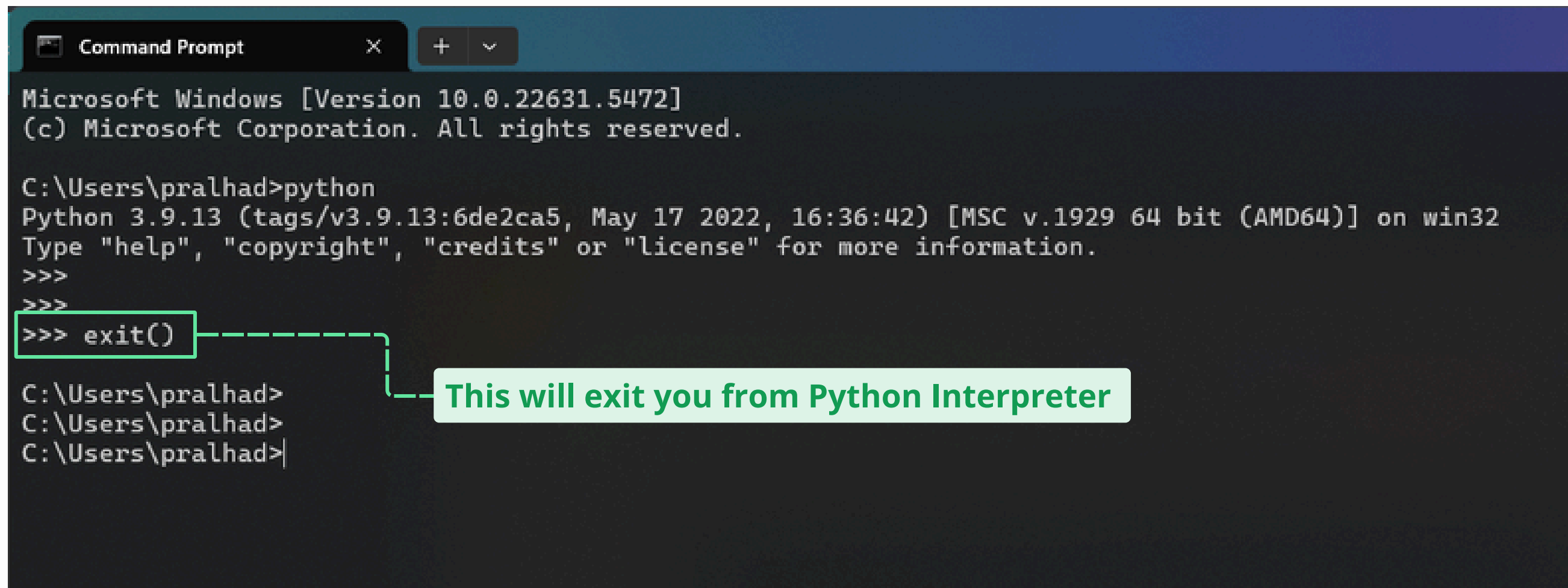
C:\Users\HP>python
Python 3.9.7 (tags/v3.9.7:1016ef3, Aug 30 2021, 20:19:38) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>>
>>>
```

where you type Python commands

- ◆ **Python Interpreter** → A program that **executes Python code**

How to Exit from Python Interpreter

👉 To exist from Python Interpreter we need to **type exit()**



```
Command Prompt
Microsoft Windows [Version 10.0.22631.5472]
(c) Microsoft Corporation. All rights reserved.

C:\Users\pralhad>python
Python 3.9.13 (tags/v3.9.13:6de2ca5, May 17 2022, 16:36:42) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>>
>>>
>>> exit()
C:\Users\pralhad>
C:\Users\pralhad>
C:\Users\pralhad>
```

A dashed green line connects the `exit()` command in the Python prompt to a green callout box.

This will exit you from Python Interpreter

Check Installed Python Version

👉 Open Command Prompt again.

- ◆ Type **python -V** and press Enter

```
python -V|
```

Command

Flag → Display Version

✅ Python is installed & ready to use! 🚀

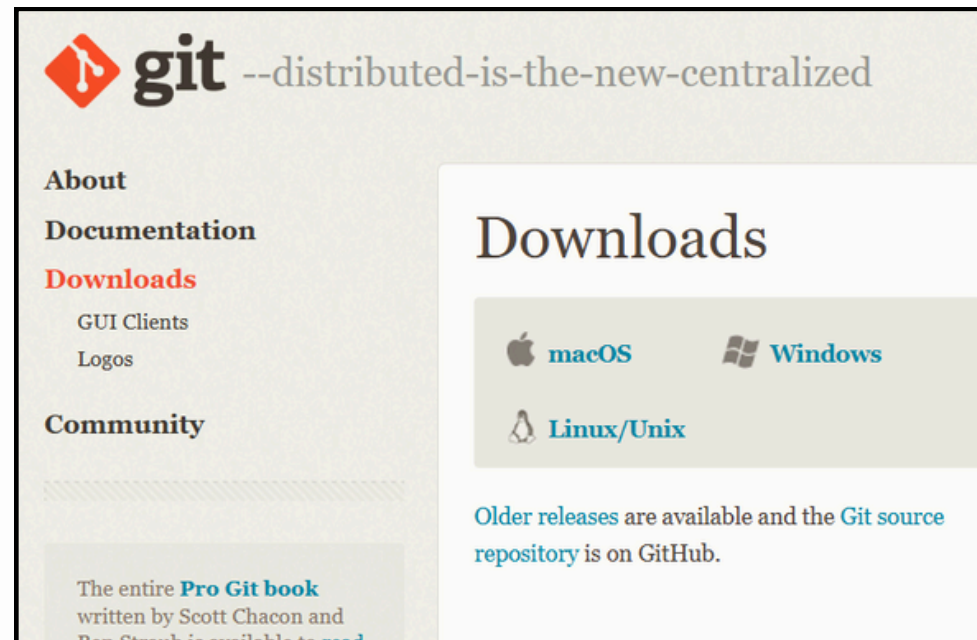
✅ Expected Output

```
Command Prompt
Microsoft Windows [Version 10.0.19045.5555]
(c) Microsoft Corporation. All rights reserved.
C:\Users\HP>python -V|
python Version
```


Git Installation

GamutX

Download Git

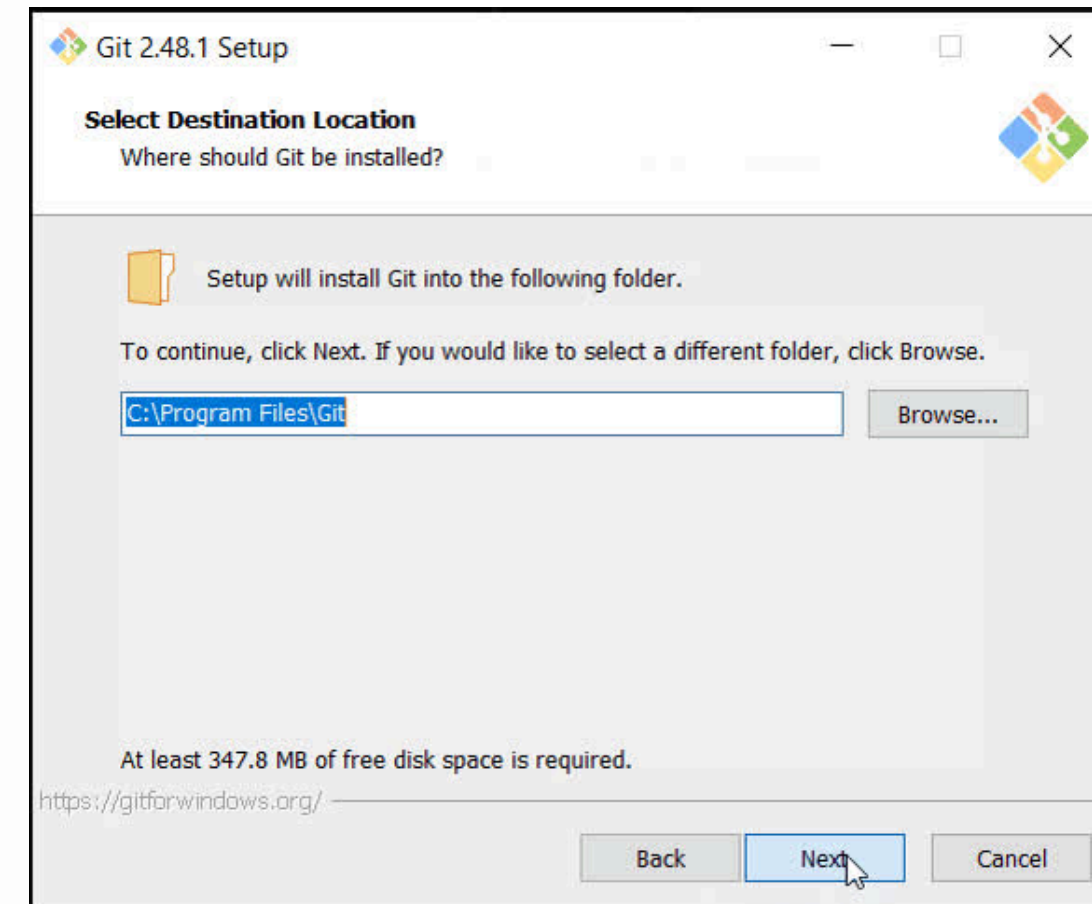


- Search "**Git Download**" on Google.
- Click on the **first link**.
- Select your **OS**
 - **Windows**
 - **Mac**
 - **Linux**

📌 Note:

- If the download doesn't start,
- select your OS and click on download link

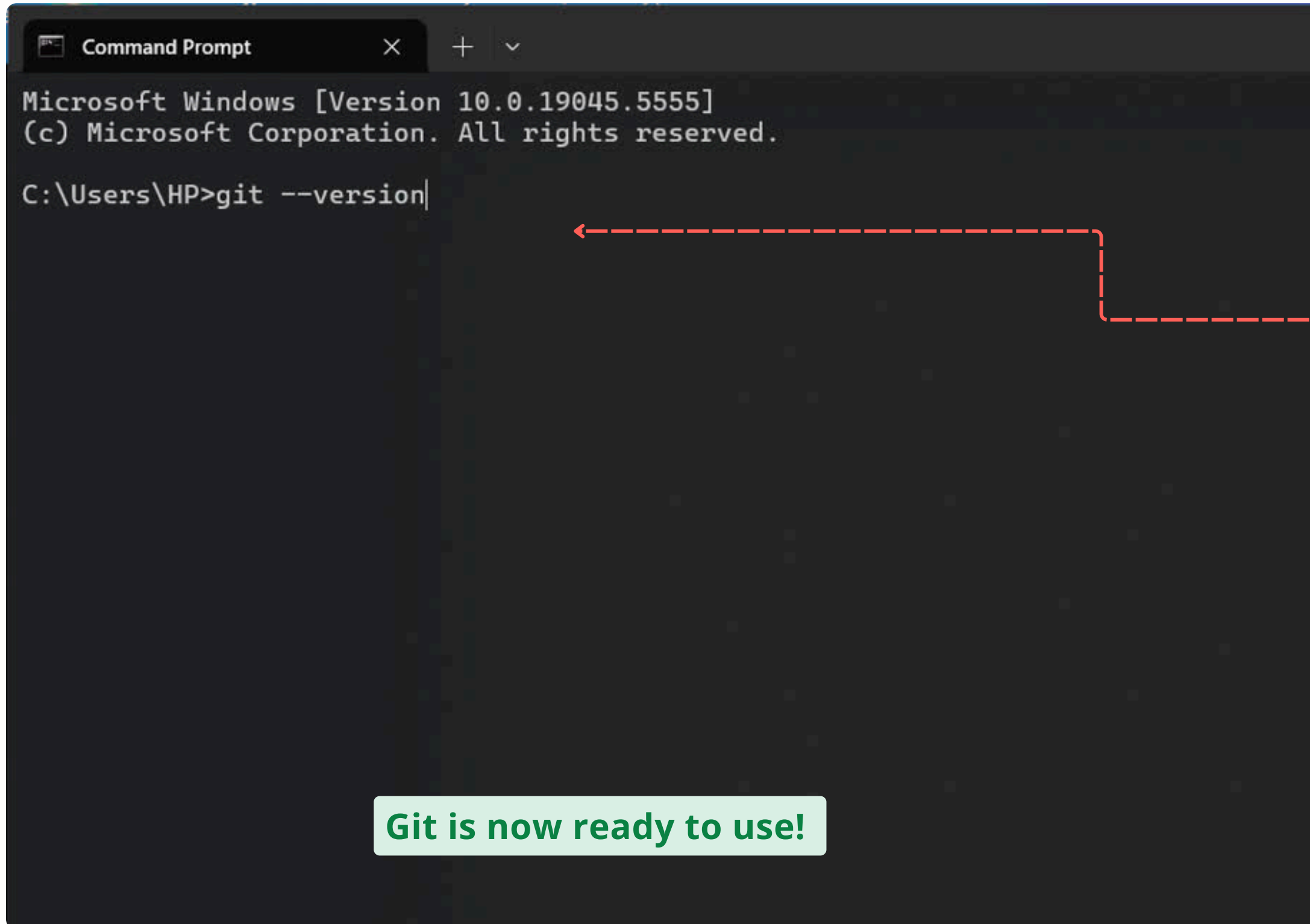
Run the Installer



- **Double Click** on git installer
- Click "**Next**" on each screen.
- Select your **preferred editor (VS Code or Vim)**.
- Keep the **default options** and click "**Install**".

🎉 **Git is now installed!** ✓

Verify the Installation



```
Microsoft Windows [Version 10.0.19045.5555]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>git --version
```

Git is now ready to use!

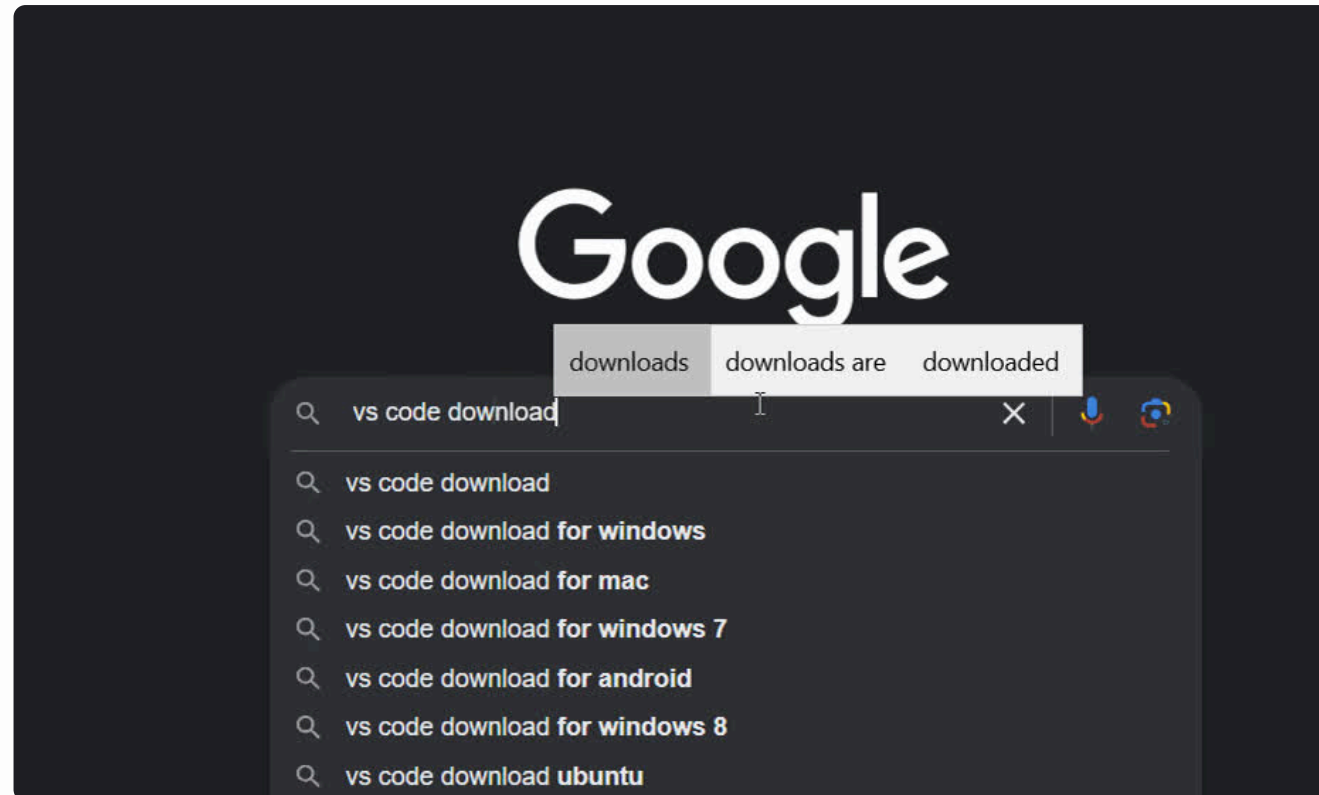
◆ Open Command Prompt

☞ Type **git --version** → Hit Enter

- If installed correctly,
- it will display the installed Git version.

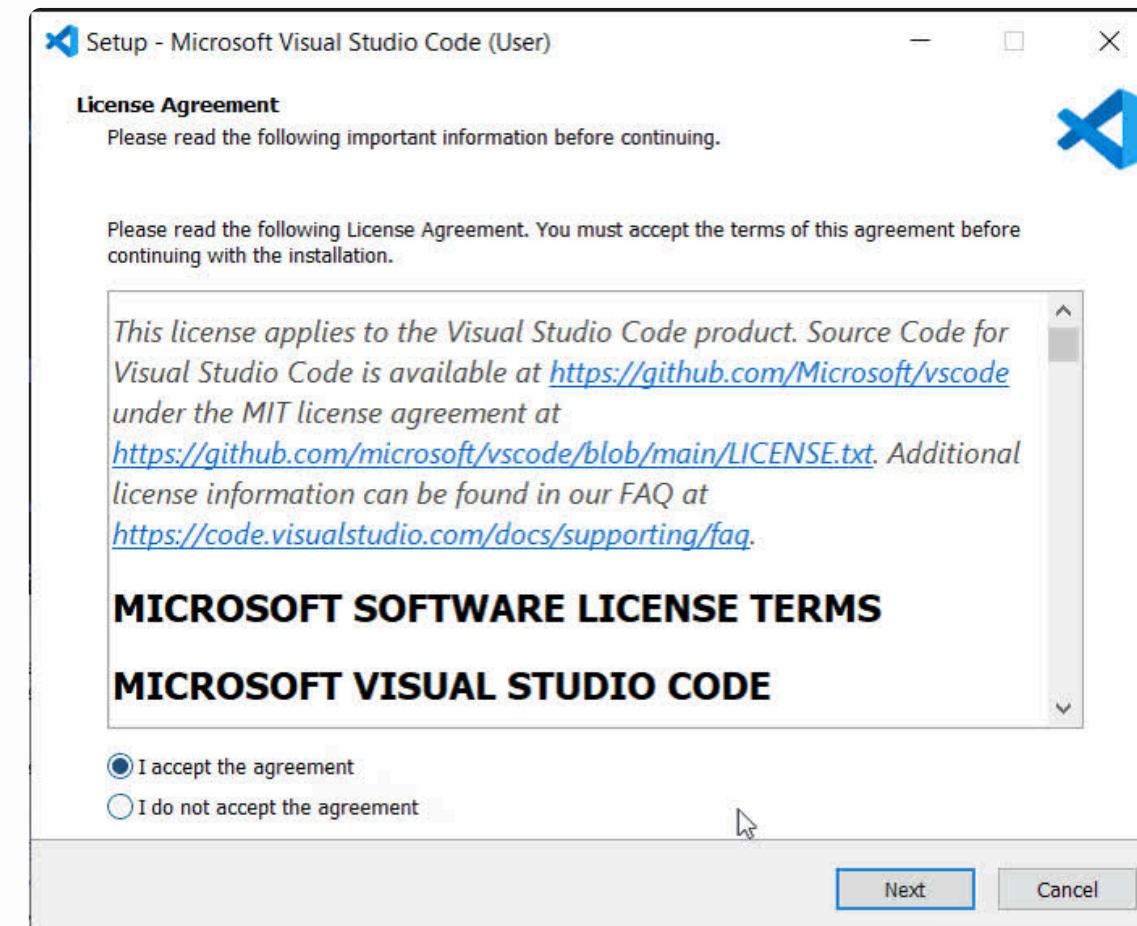
IDE Installation (VS Code)

Download VS Code



- Search “VS Code Download” on Google
- Click on **First Link**
- Click the **Download** button for your OS:
 - 🖥️ **Windows** , 🍏 **macOS** , 🐧 **Linux**

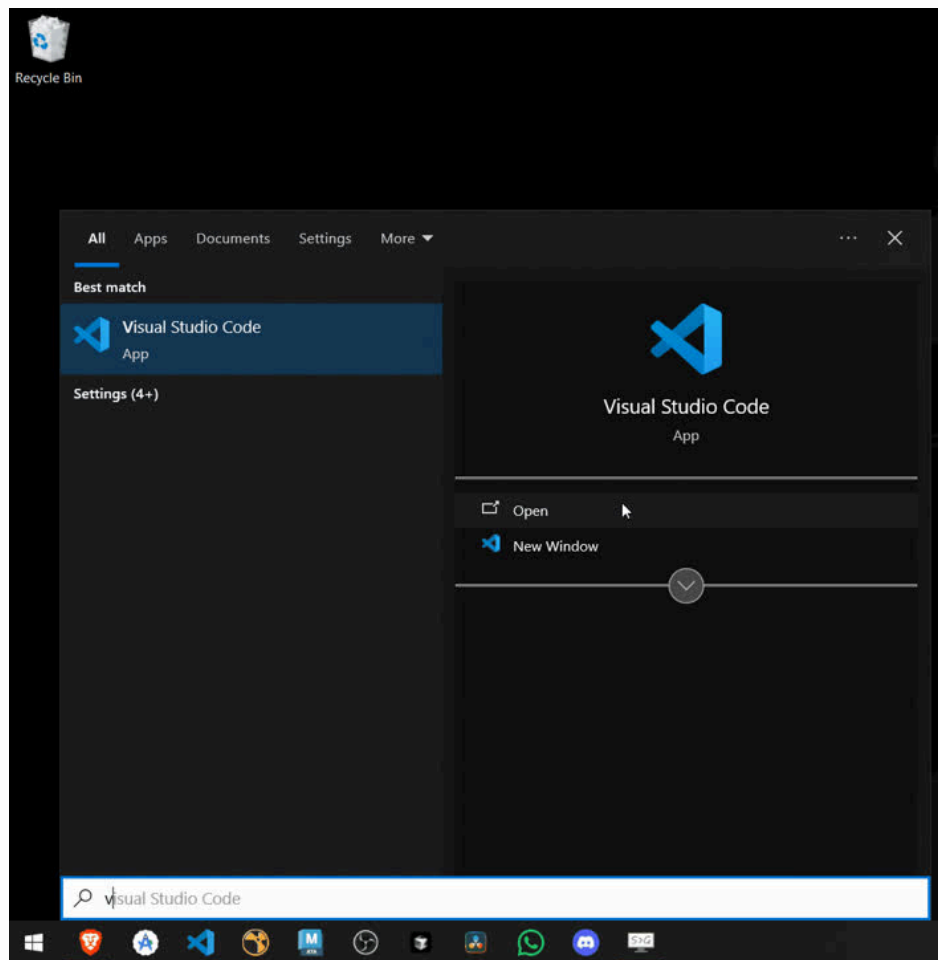
Run the Installer



- **Double Click** on Vs code Installer.
- **Accept the agreement** & click Next.
- Choose the **installation location** & click Next.
- Select **additional options**
 - ➡ **Open with Code** & **Add to PATH**
- Click **Install** & wait for completion.

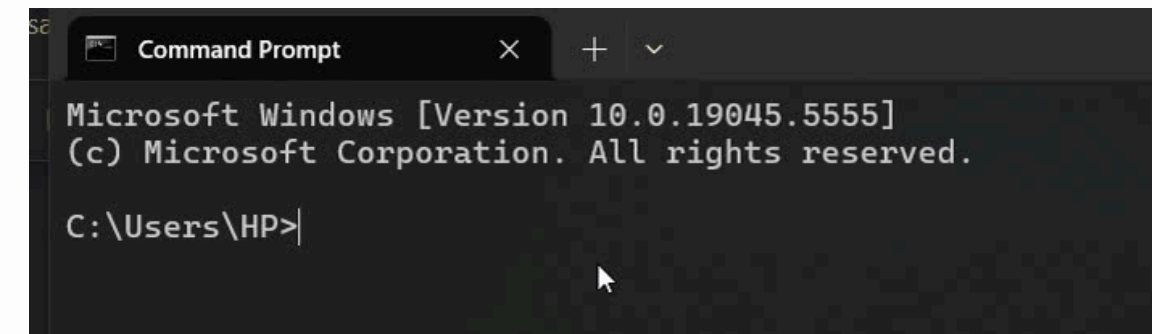
Verify the Installation

Open from Start Menu



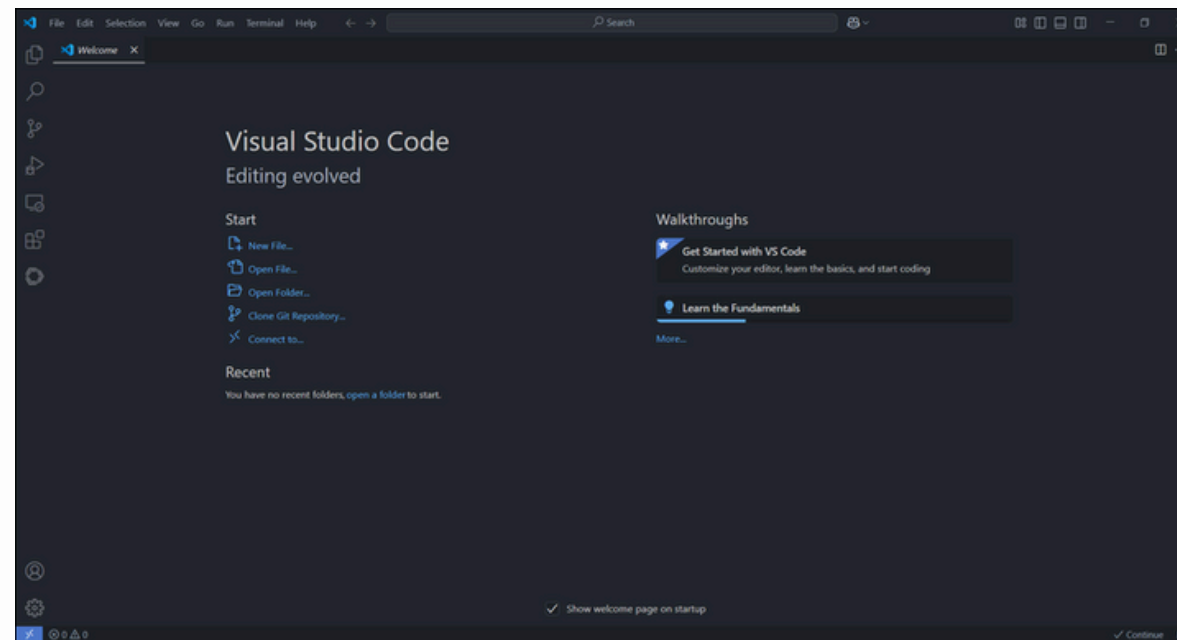
Open from Command Prompt

◆ Open Command Prompt



👉 Type **code** → Hit Enter

VS Code is now installed and ready to use.



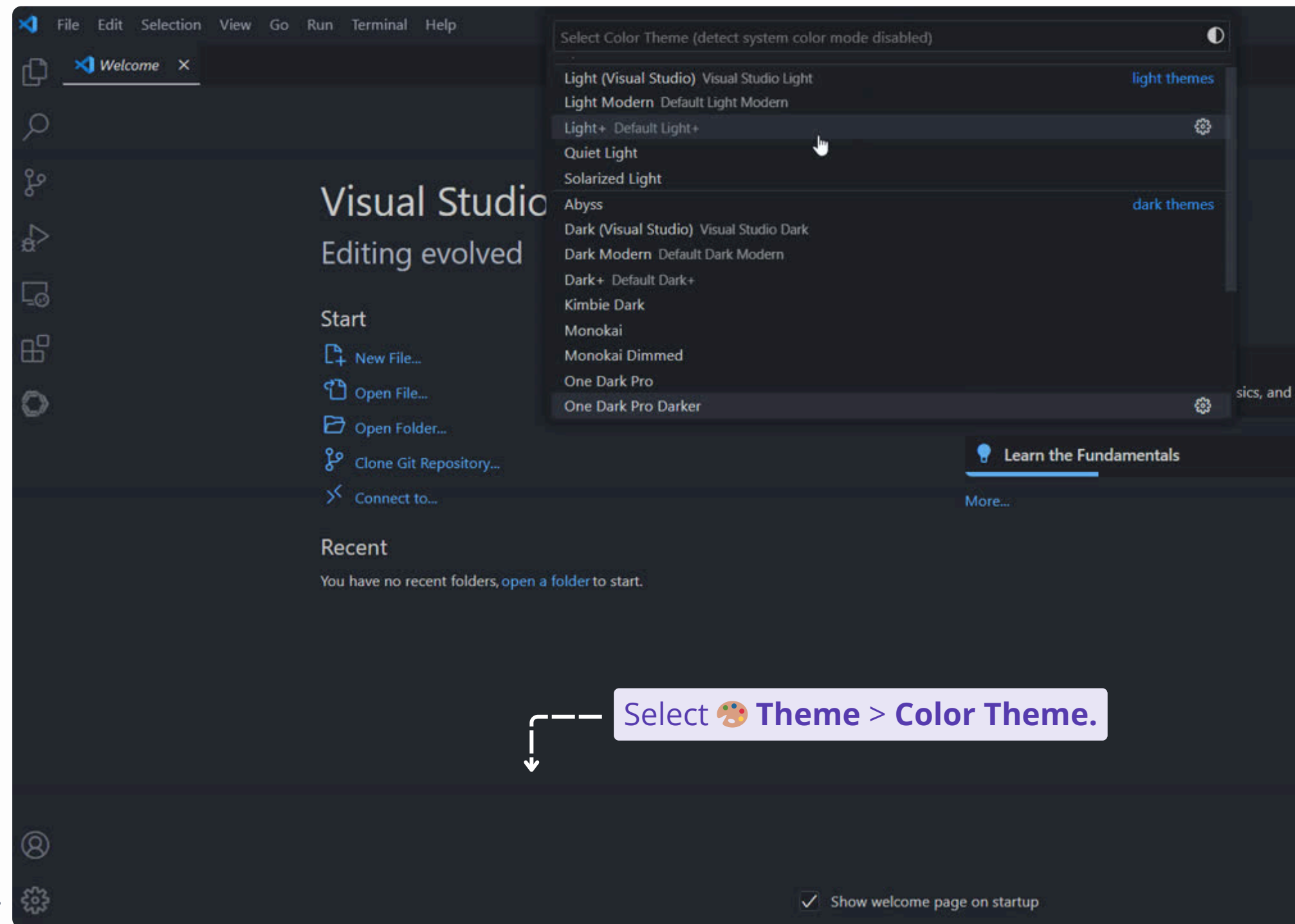
OUTPUT

OUTPUT

- Click on the **Start Menu**.
- Search for "**Visual Studio Code**" and open it.

Configuring Visual Studio Code

Change Theme

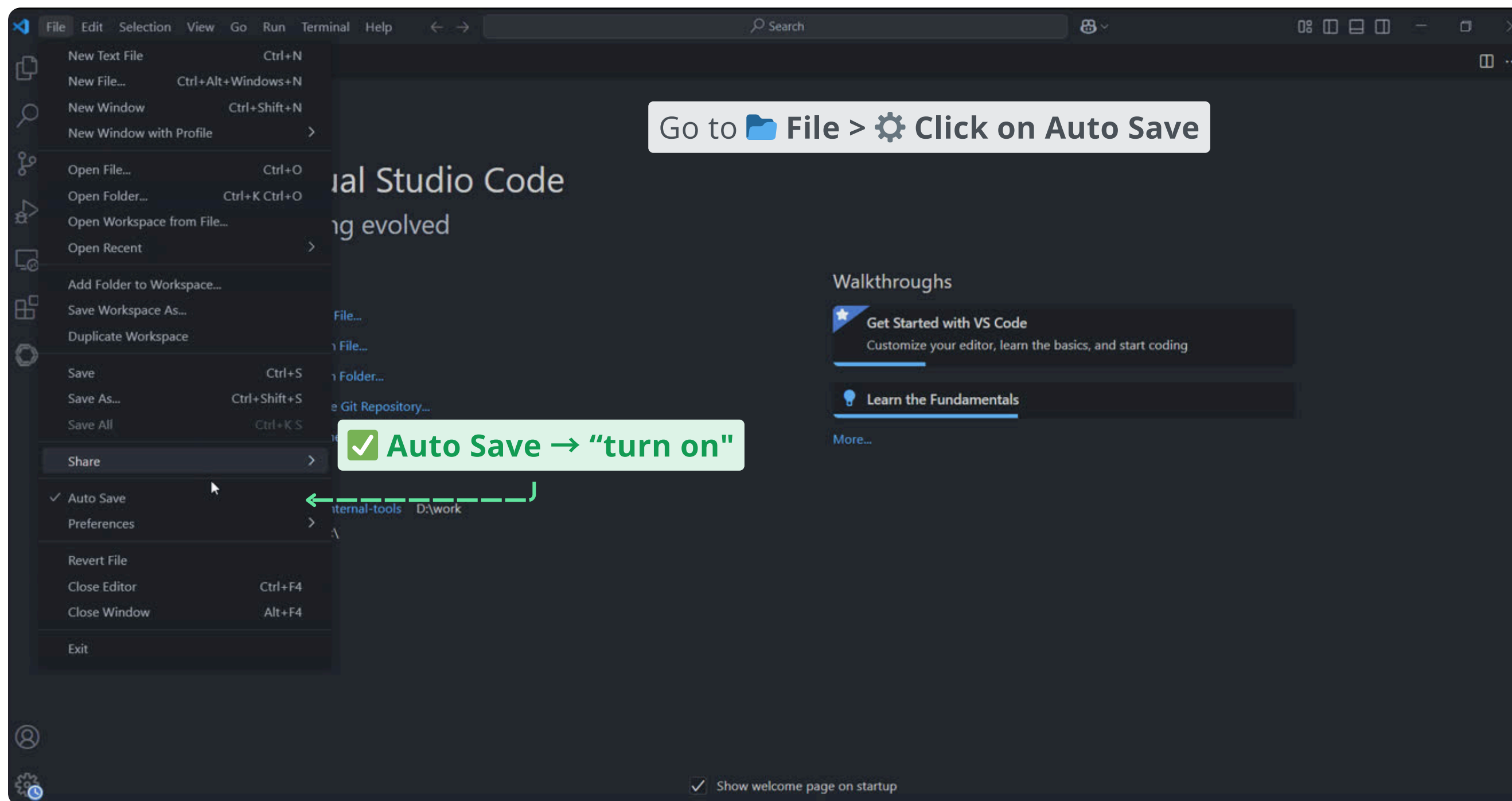


Choose your **preferred theme.**

Click the ⚙️ **Gear icon**

Select 🎨 **Theme > Color Theme.**

Enable Autosave

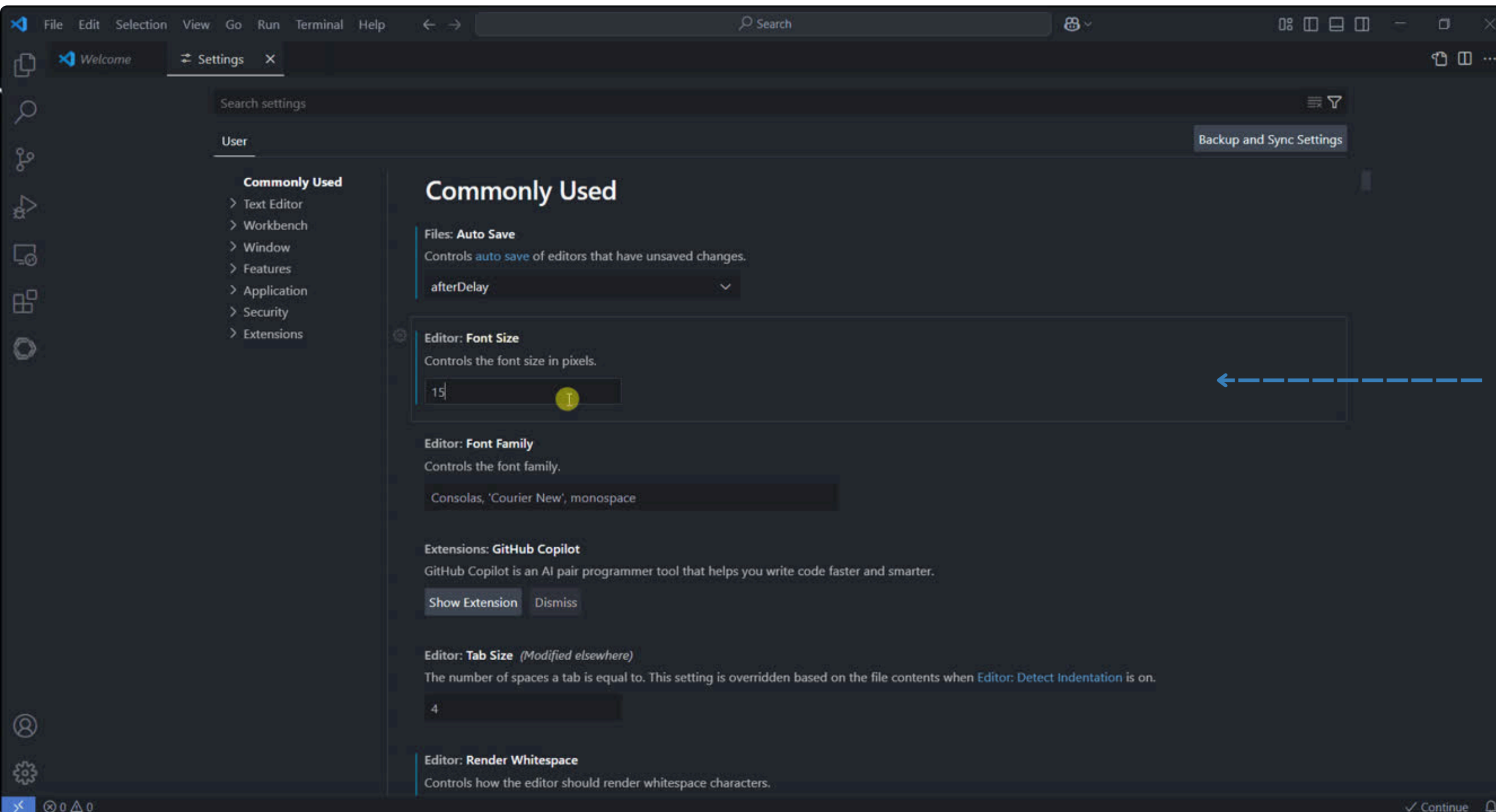


📌 Auto-saves work to **prevent data loss**.

Adjust font size

GamutX

Go to  File >  Preferences >  Settings.



 Editor: **Font Size** → Adjust **font size**.

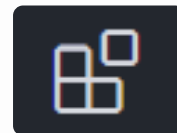
 Improves **readability** and enhances coding comfort.

Installing Initial Extensions

- Extensions enhance functionality in VS Code, making **development easier and more efficient**.



Open VS Code



Click Extensions in the Sidebar

OR press **Ctrl + Shift + X**

Python Extension

Search for **Python**.



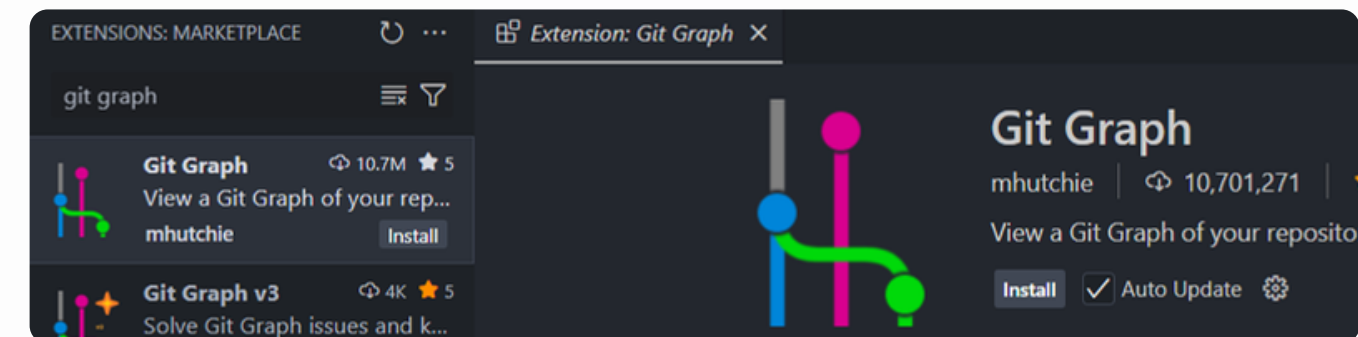
✓ Why it's important?

- Enables Python **auto-completion** & **suggestions**
- Provides **Linting** & **Debugging tools**.



Git Graph Extension

Search for **Git Graph**.



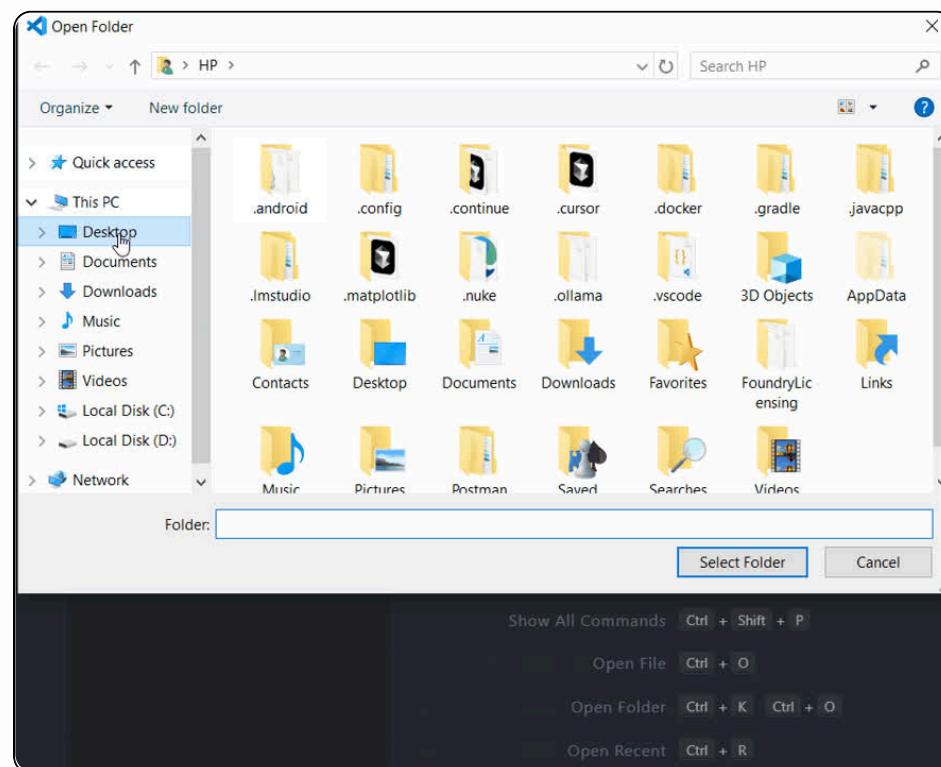
✓ Why it's important?

- Visualizes Git commit history** in a simple graph.
- Helps **track changes** without using the terminal.

Opening a Project/Folder in VS Code

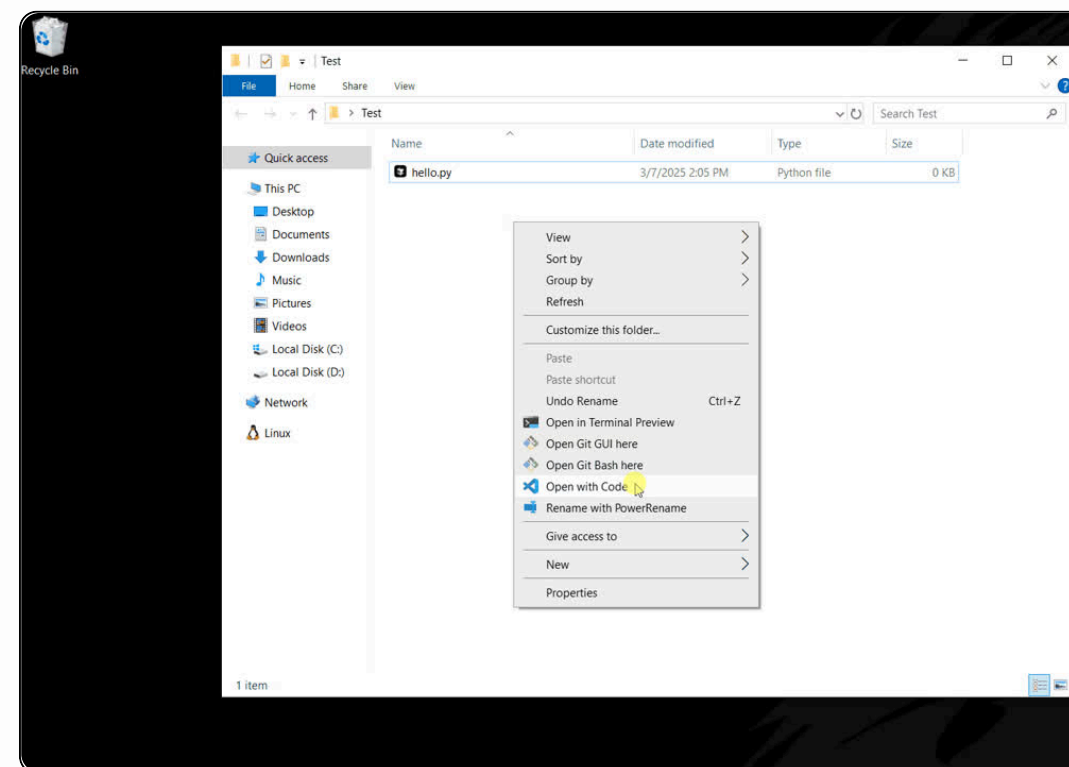
- There are **three ways** to open a **folder** in VS Code:

Open Directly in VS Code



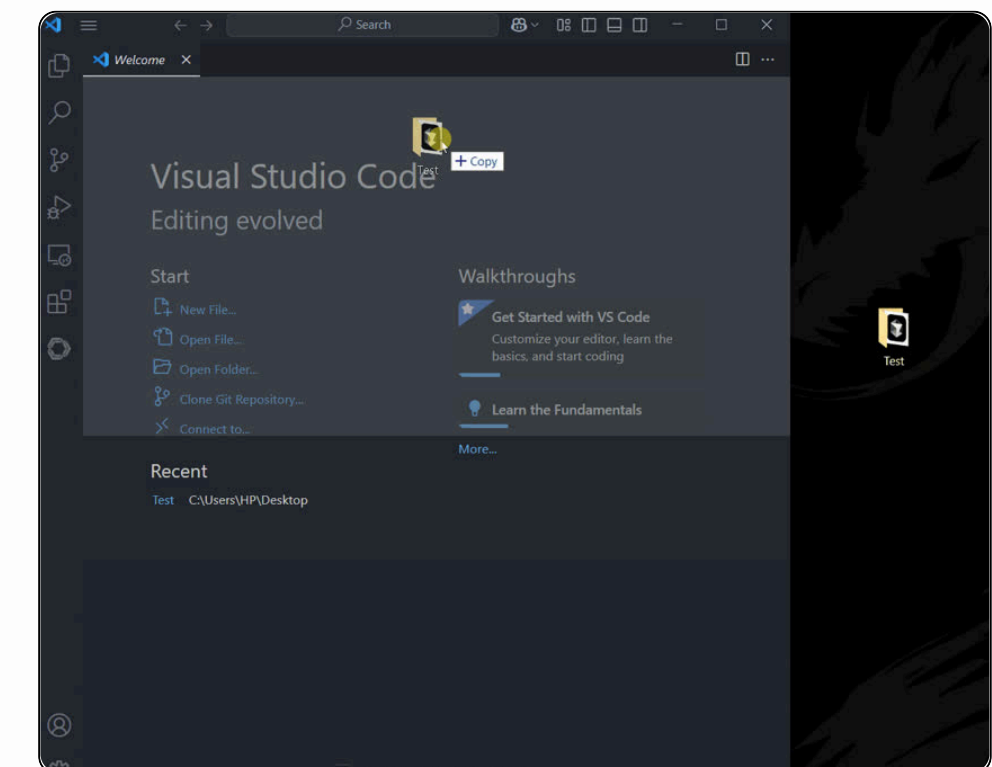
- Create a **new folder** name **Test**
- Open **VS Code**.
- Click **File > Open Folder..**
- Select the folder and click **Select Folder**

Open via Right-Click



- Navigate to the **folder** in File Explorer.
- Right-click** on the folder.
- Select "**Open with Code**" from menu.
- VS Code **launches** with the **folder open**.

Drag & Drop



- Locate the folder** you want to open.
- Drag & Drop** the folder into the **VS Code window**.
- The folder opens instantly!

Note: We are using the same folder for all these methods

Running Your First Python File in VS Code

- Creating Python Files
- VS Code Terminal
- Running Python File in VsCode Terminal Manually
- Understanding CMD & PowerShell
- Running Python Files in PowerShell
- What Does . (Dot) Mean in PowerShell?
- What is File Path Autocomplete?
- Understanding Relative Path vs. Absolute Path

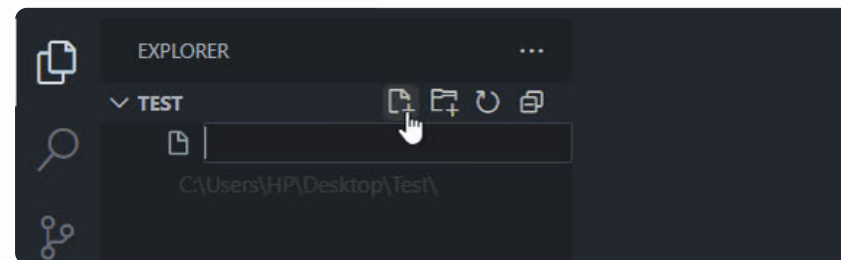
Creating Python Files

Creating a Python File

 Open VS Code

 Open your Project Folder

👉 Explorer panel, click New File



or press **Ctrl + N**

hello.py

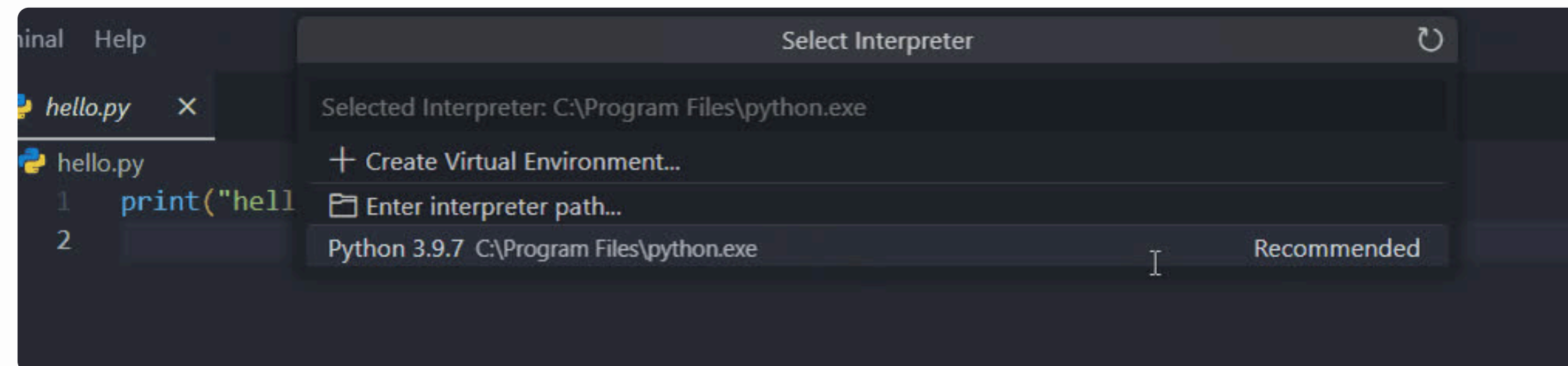
✅ Save the file with a **.py** extension

Writing Code

```
print("Hello, World !")
```

◆ Write same code in **hello.py** File

Selecting the Python Interpreter

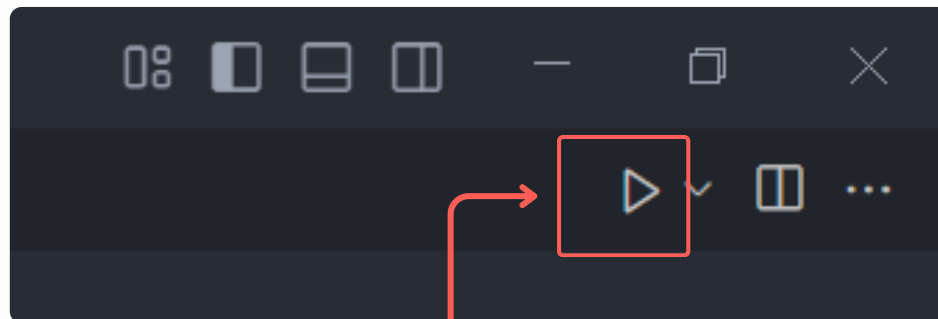


Why select a Python interpreter?

- ✅ Ensures VS Code runs
 - **correct Python version.**

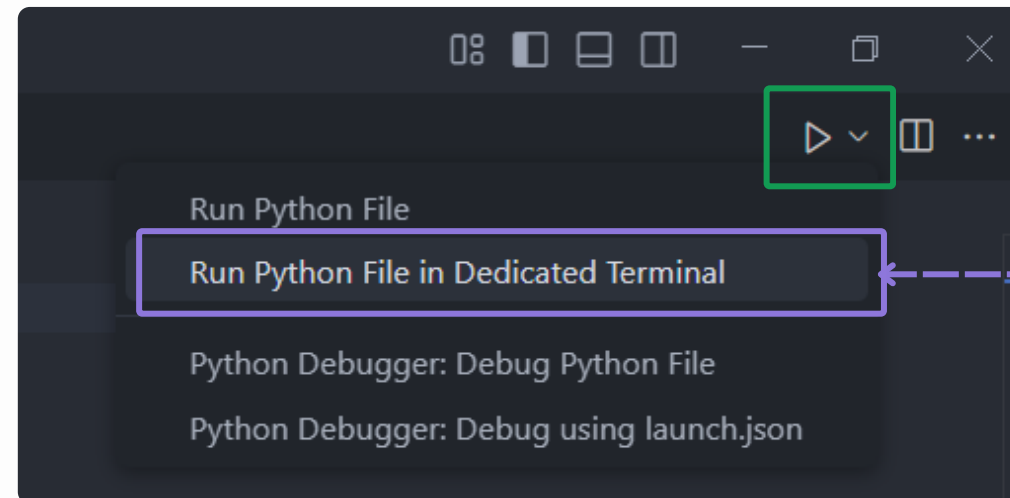
- Open the **Command Palette** by pressing **Ctrl + Shift + P** Type "**Python Select Interpreter**" and click it.
- Select the **Recommended Python interpreter** from the list.

Running a Python File



Click the **Run button (▶)**

Alternative way



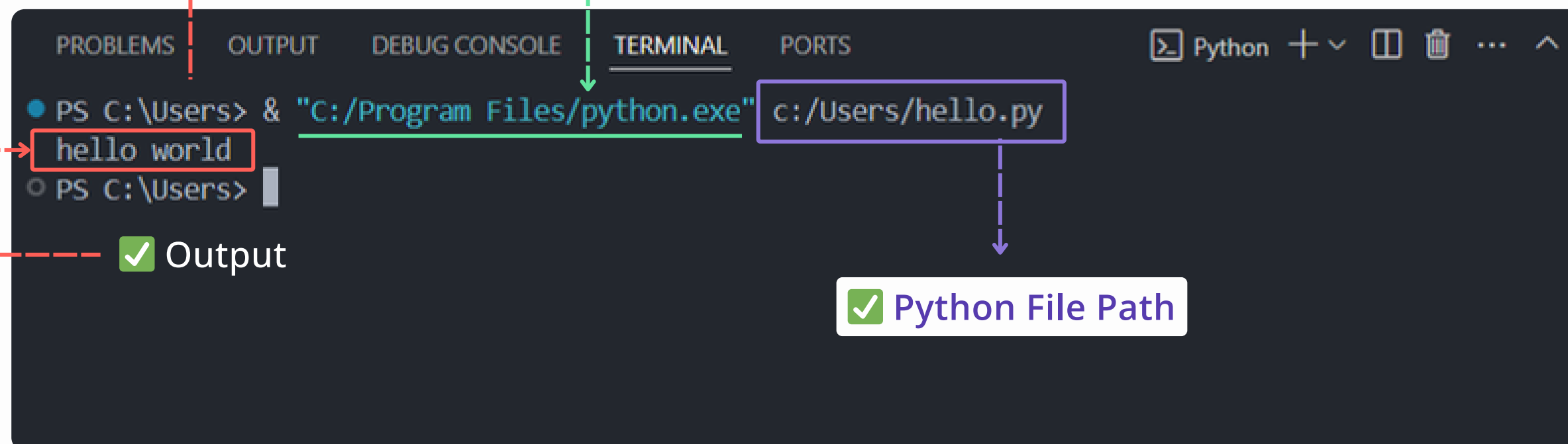
Click on **Down Arrow**

Select "**Run Python File in Dedicated Terminal**"

 **Output**

 **Current Directory**

Python Path

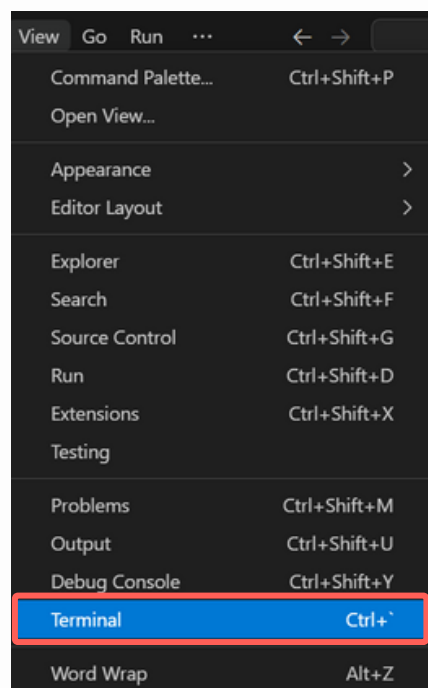


 **Python File Path**

VS Code Terminal

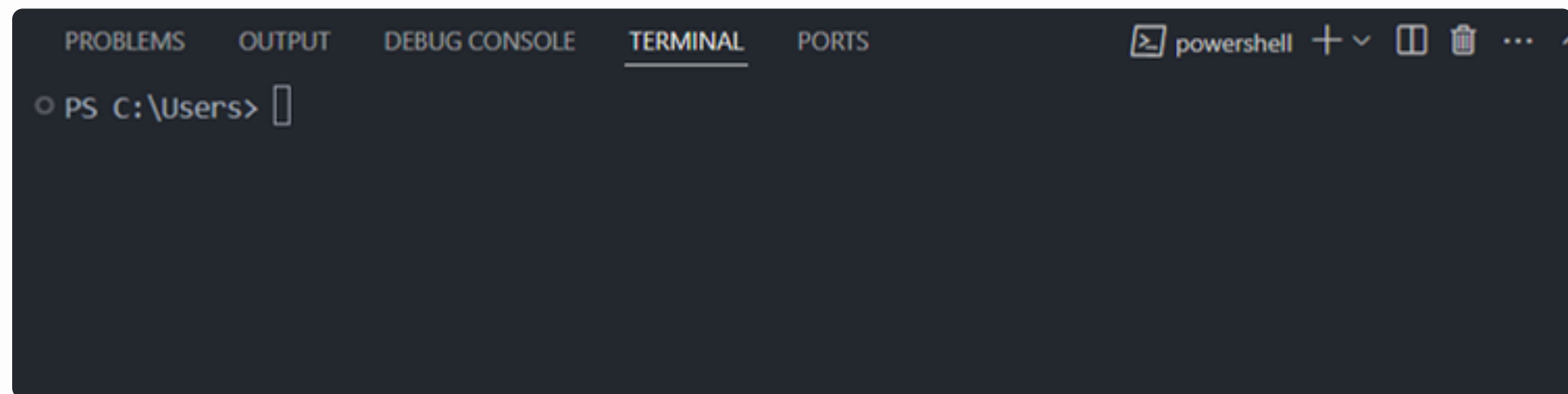
Learn how to efficiently **open, close, and manage multiple terminals** in VS Code

Opening the Terminal

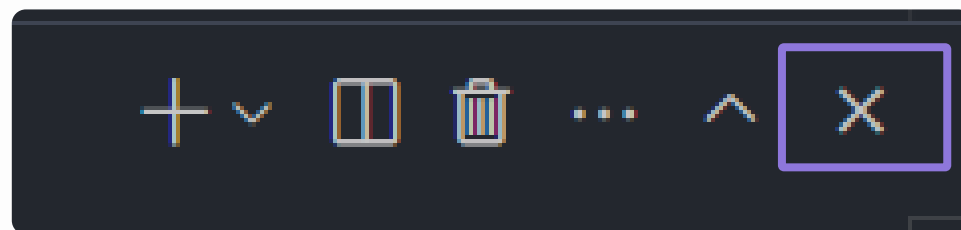


View > Terminal or
use **Ctrl+`** (Backtick)

Terminal



Closing the Terminal

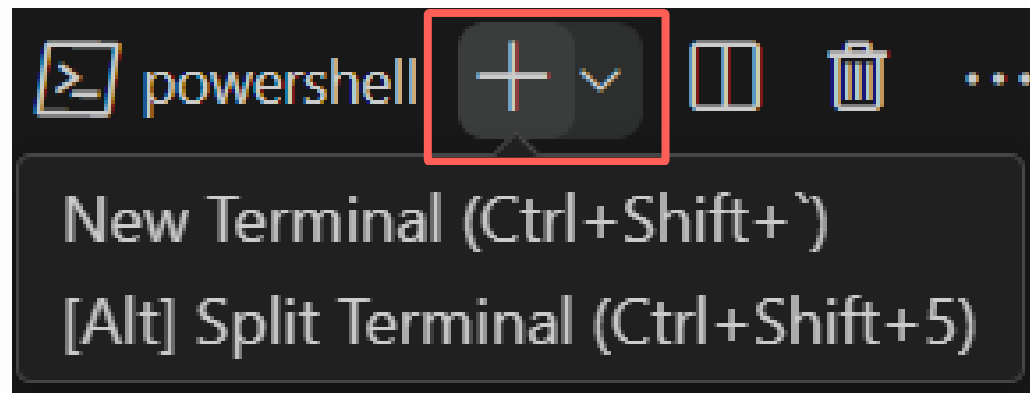


✓ Click the "X" icon → **closes the terminal tab**

- ✓ Keeps your **workspace clean** and organized.
- ✓ **Prevents confusion** when managing **multiple tasks**.

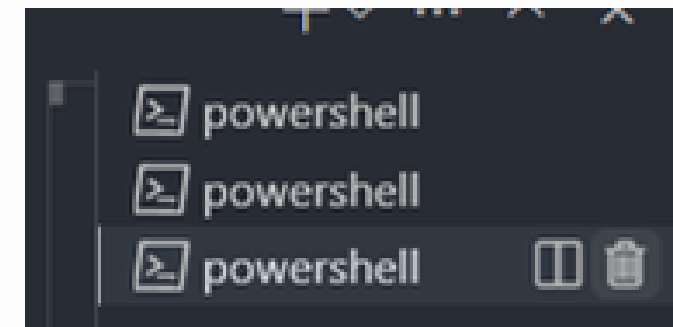
NOTE: Use Ctrl + ` ---> hides/shows the terminal

Managing Multiple Terminals



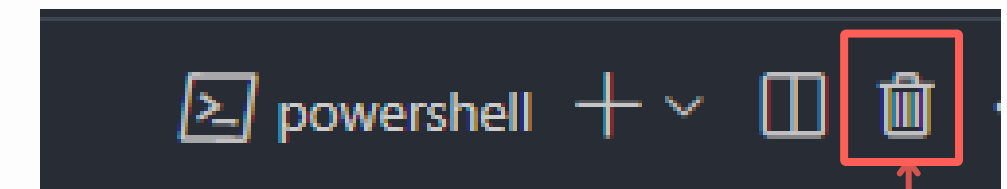
✓ Click the **+** plus icon . or press **Ctrl + Shift + `**

✓ Switch Between Terminals



Click on the **specific terminal name** from the list.

✗ Closing the Terminal



Click **closes the selected terminal**

Why Managing Terminals is Useful?

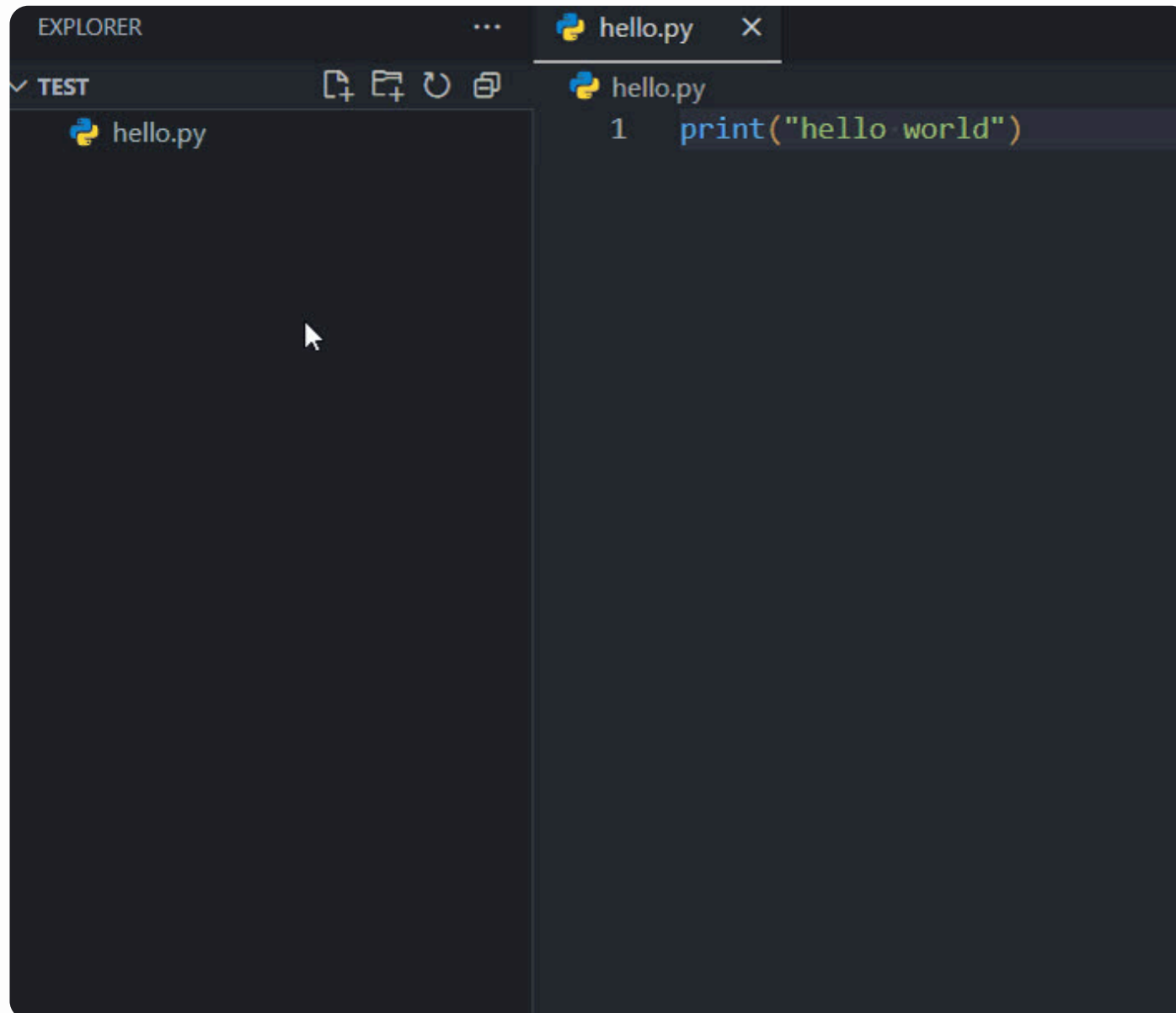
- Work on **different tasks** at the **same time**.
 - Switch between terminals **without reopening them**.

✓ Key Point to Remember

- ✓ Click the trash icon 🗑️
closes the selected terminal
- ✓ Use Ctrl + `
hides/shows the terminal
- ✓ Click the "X" icon
closes the terminal tab

Running Python File in VsCode Terminal Manually

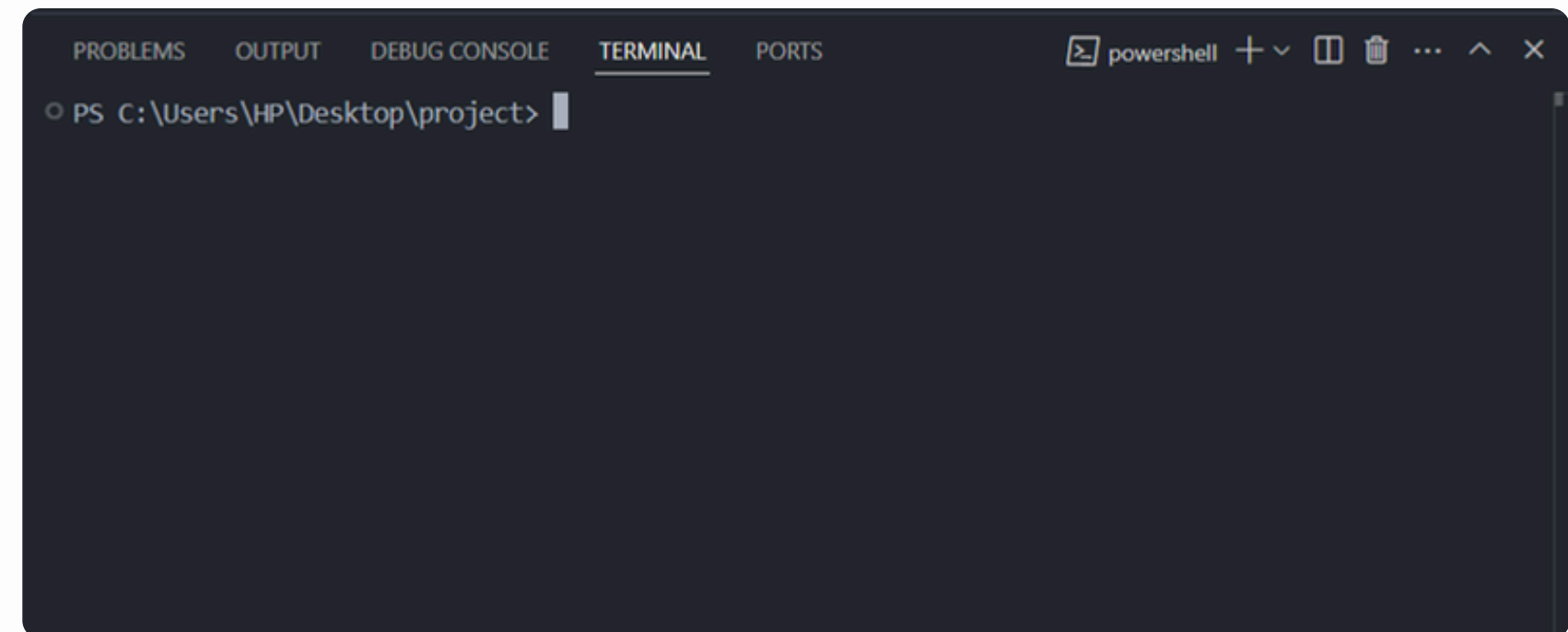
Copy the Path



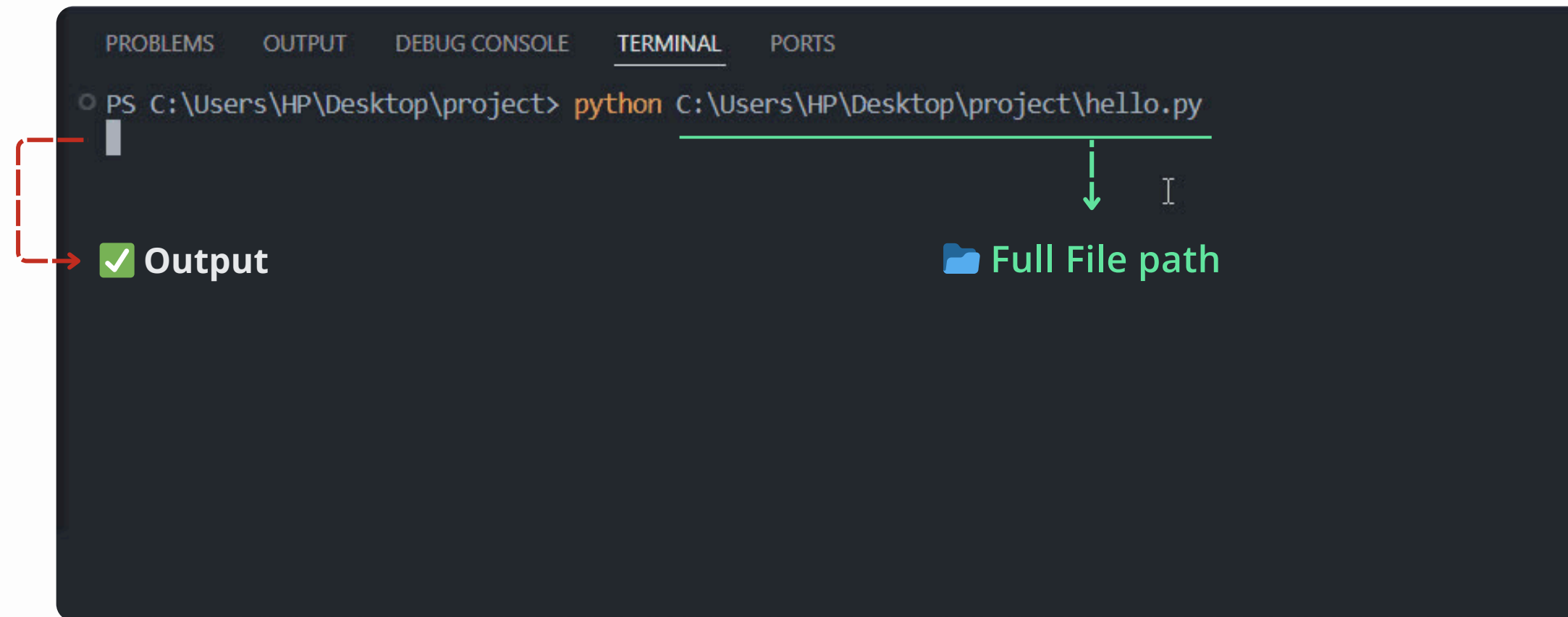
- In VS Code Explorer
- **Right-Click** on the **hello.py** file.
- Select "**Copy Path**".

Open VS Code Terminal

Press **Ctrl + `** (**Backtick**)



Paste the copied path



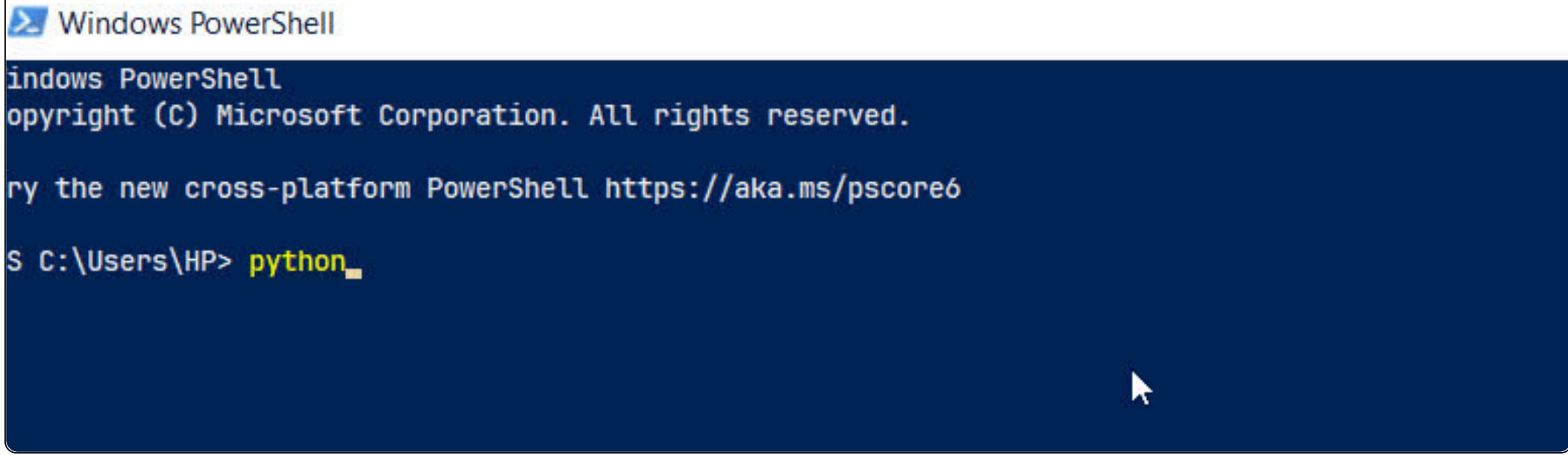
The screenshot shows a terminal window with tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The TERMINAL tab is active, displaying a command prompt window with the command `python C:\Users\HP\Desktop\project\hello.py`. A green arrow points from the file path `C:\Users\HP\Desktop\project\hello.py` to a folder icon labeled "Full File path". A red dashed box highlights the command prompt area, and a green checkmark icon is next to the word "Output".

Why Run a Python File in the Terminal?

- Run Python **without an editor**.
- Execute from **any location**
- **Quickly test scripts** without opening an IDE.

Verify Git & Python installation in Powershell

Check Python installation



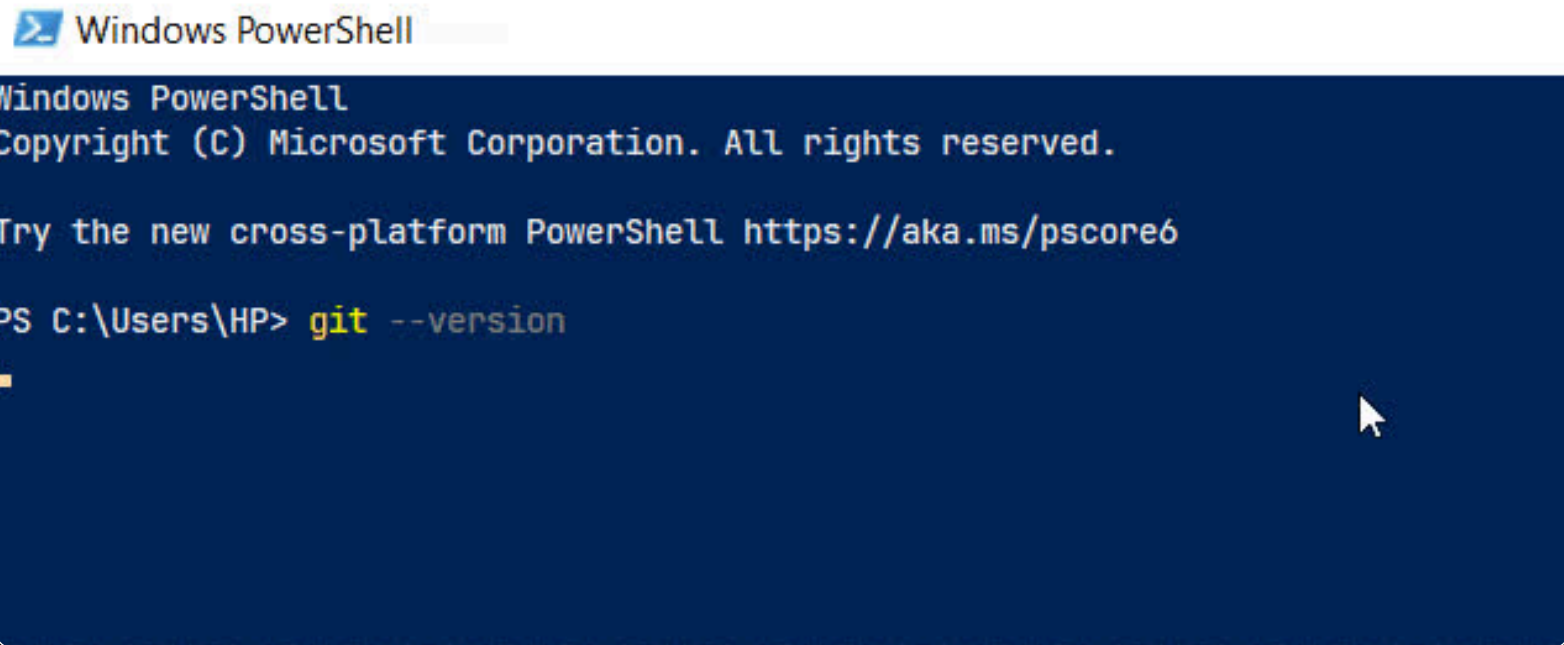
```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\HP> python_
```

Open PowerShell and Type “**python**”

Check Git Installation



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\HP> git --version
```

Open PowerShell and Type “**git --version**”

Running Python Files in PowerShell

Two Ways to Run Python Files in PowerShell are

- **Run from Anywhere –**
 - Execute the Python file **without changing directories.**
- **Run After Navigating To Script Directory –**
 - Navigate to the file **location first, then execute it.**

Run Python File from Anywhere

Script Location:

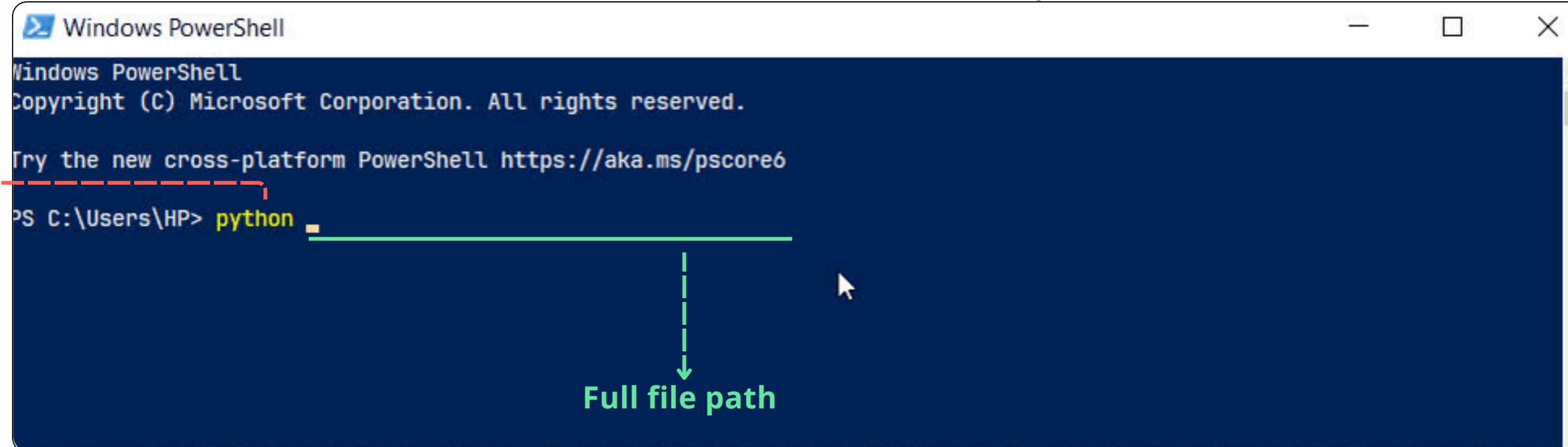
"C:\Users\HP\Desktop\project\hello.py"

How to do it ?

Open **PowerShell**
Run the command

◆ Execute Python scripts

- instantly from any location **using the full file path.**



The screenshot shows a Windows PowerShell window with a dark blue background. The title bar reads "Windows PowerShell". The terminal text includes the copyright notice for Microsoft Corporation and a link to the new cross-platform PowerShell. The command prompt shows "PS C:\Users\HP> python" followed by a green horizontal line representing the full file path. A green dashed arrow points from the text "Full file path" below the terminal to the green line. A red dashed arrow points from the text "Calls Python interpreter" below the terminal to the word "python" in the command.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\HP> python
```

Full file path

Calls Python interpreter

Run After Navigating To Script Directory

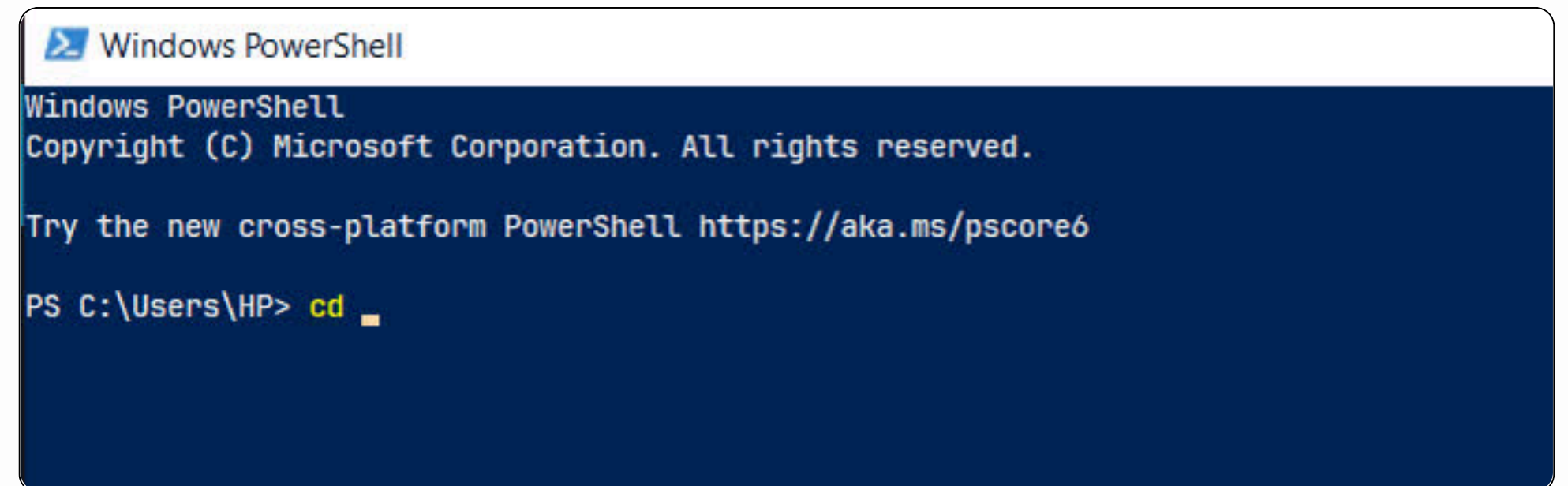
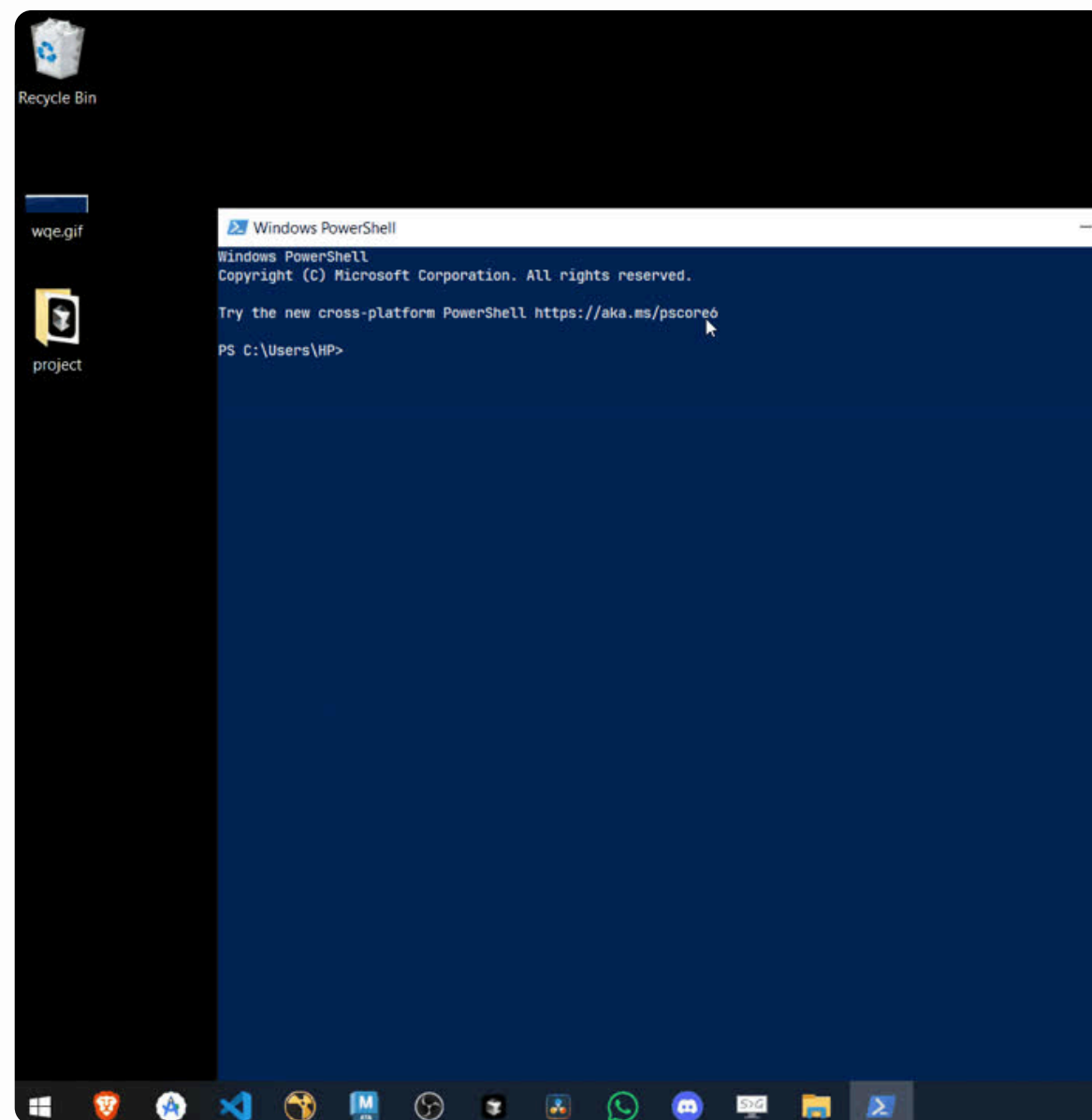
📌 **Target Folder:** C:\Users\HP\Desktop\project

Open PowerShell

✓ Click on Start and search **PowerShell**

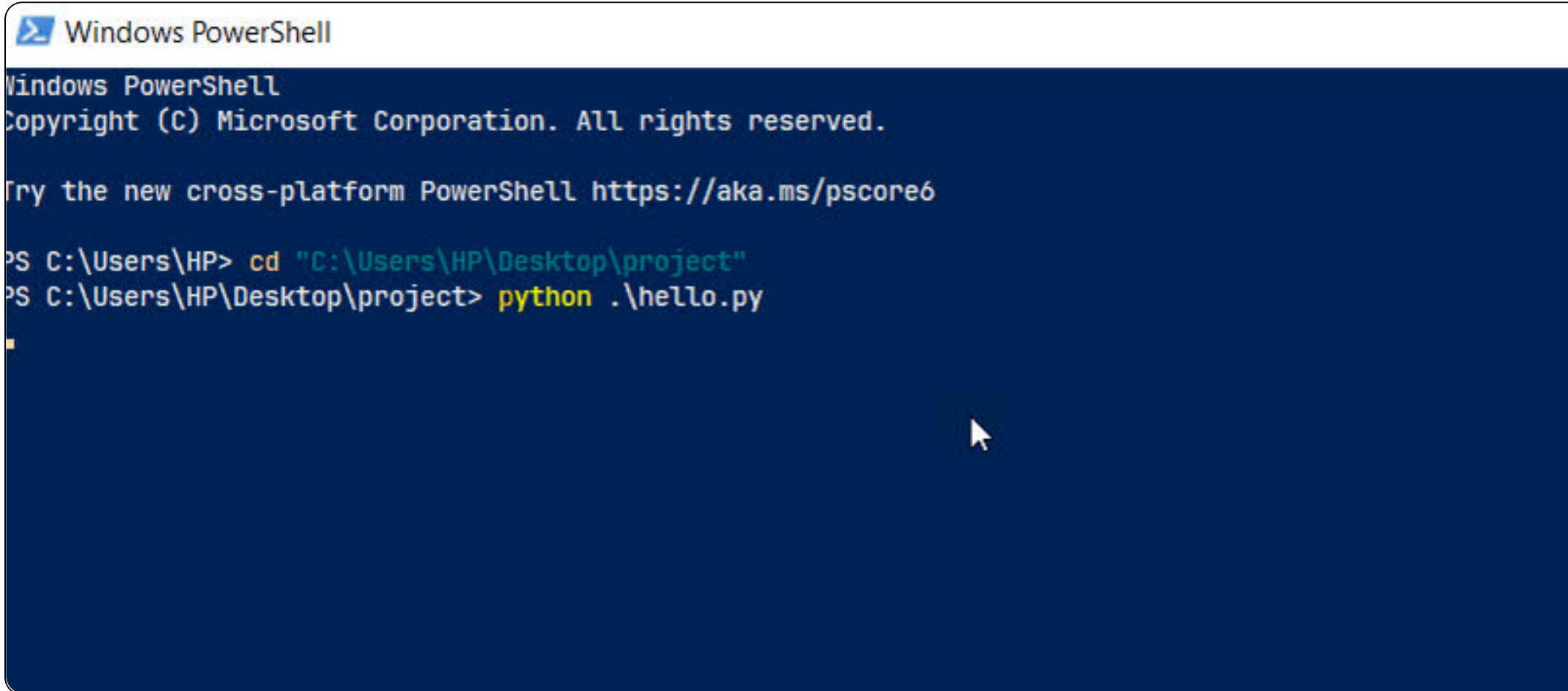
📁 **Current Location:** C:\Users\HP

Type --> **cd "C:\Users\HP\Desktop\project"**



cd = Change Directory

Run the Python Script



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\HP> cd "C:\Users\HP\Desktop\project"
PS C:\Users\HP\Desktop\project> python .\hello.py
```

✓ To run file

- ➡ Type : **python .\hello.py**

What's the use of it ?

- ✓ **No need** to type the **full file path**.
- ✓ Python will find and run **hello.py** in the **current folder**.

What Does . (Dot) Mean in PowerShell?

- Represents the **current directory**
- A **shorthand** to indicate where we are working.
- Useful for **referencing relative paths**

Use Cases of (.)

Access Files in the **Current Directory**



Run a file as

```
python .\hello.py
```

What it means:

- . (dot) = Current folder
- .\hello.py = "It's in the current directory"

✓ Why is this useful?

- **Saves time** –
No need to type long paths!
- **Works for all files Types** –
Scripts, text files, etc.
- **Simplifies navigation** –
Easily access files in your folder.
- **Essential for scripting** –
Useful for automation and file handling.

What is File Path Autocomplete?

💡 What is it ?

- A PowerShell feature that auto-fills file/folder paths
- It completes paths as you type

How to Use File Path Autocomplete

- ◆ If your file path is → "C:\Users\hello.py"

Start typing `python C:\Us` ← in powershell

```
python C:\Us
```

Press the Tab key to autocomplete.

```
python C:\Users\
```

Press Tab repeatedly to cycle through options

```
python "C:\Users\hello.py"
```

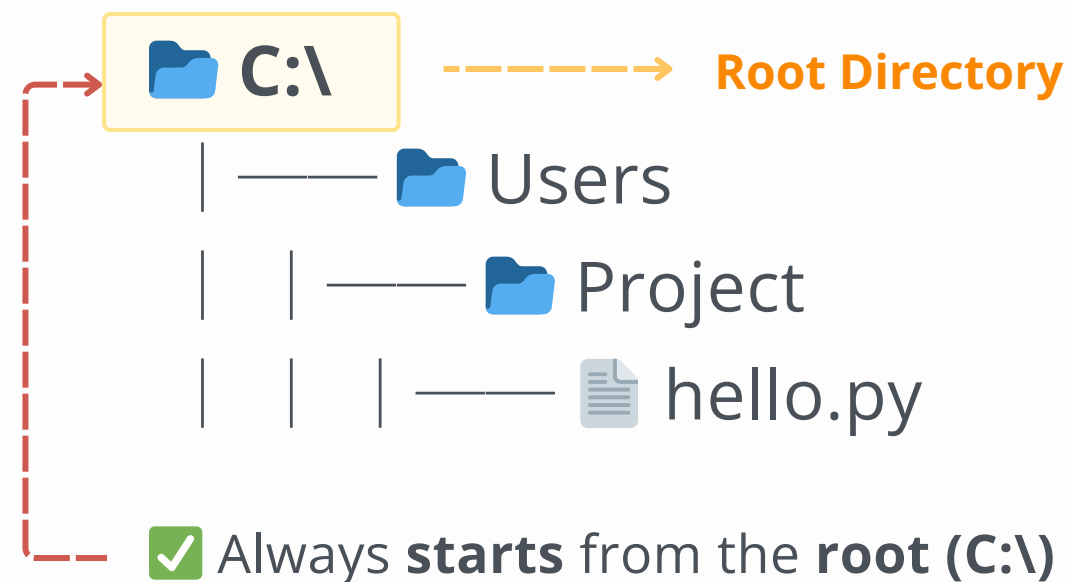
✅ Why Do We Use It?

- **Saves Time** – No need to type long file paths manually.
- **Prevents Errors** – Ensures accurate file/folder names.

Understanding Relative Path vs. Absolute Path

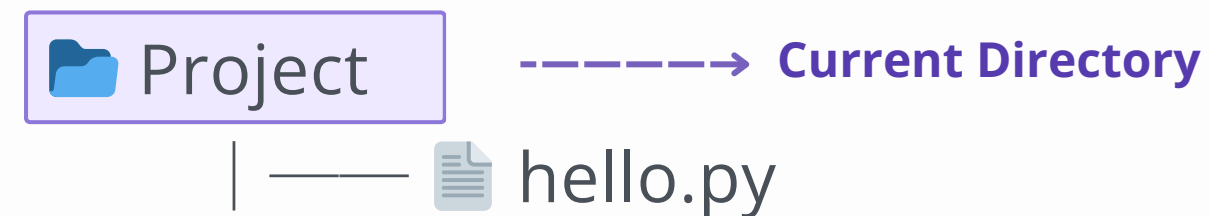
Absolute Path

Path → `C:\Users\Project\hello.py`



Relative Path

Path → `Project\hello.py`



✓ Depends on the **current directory**.

Which One is Preferred?

💡 Use Absolute Path if:

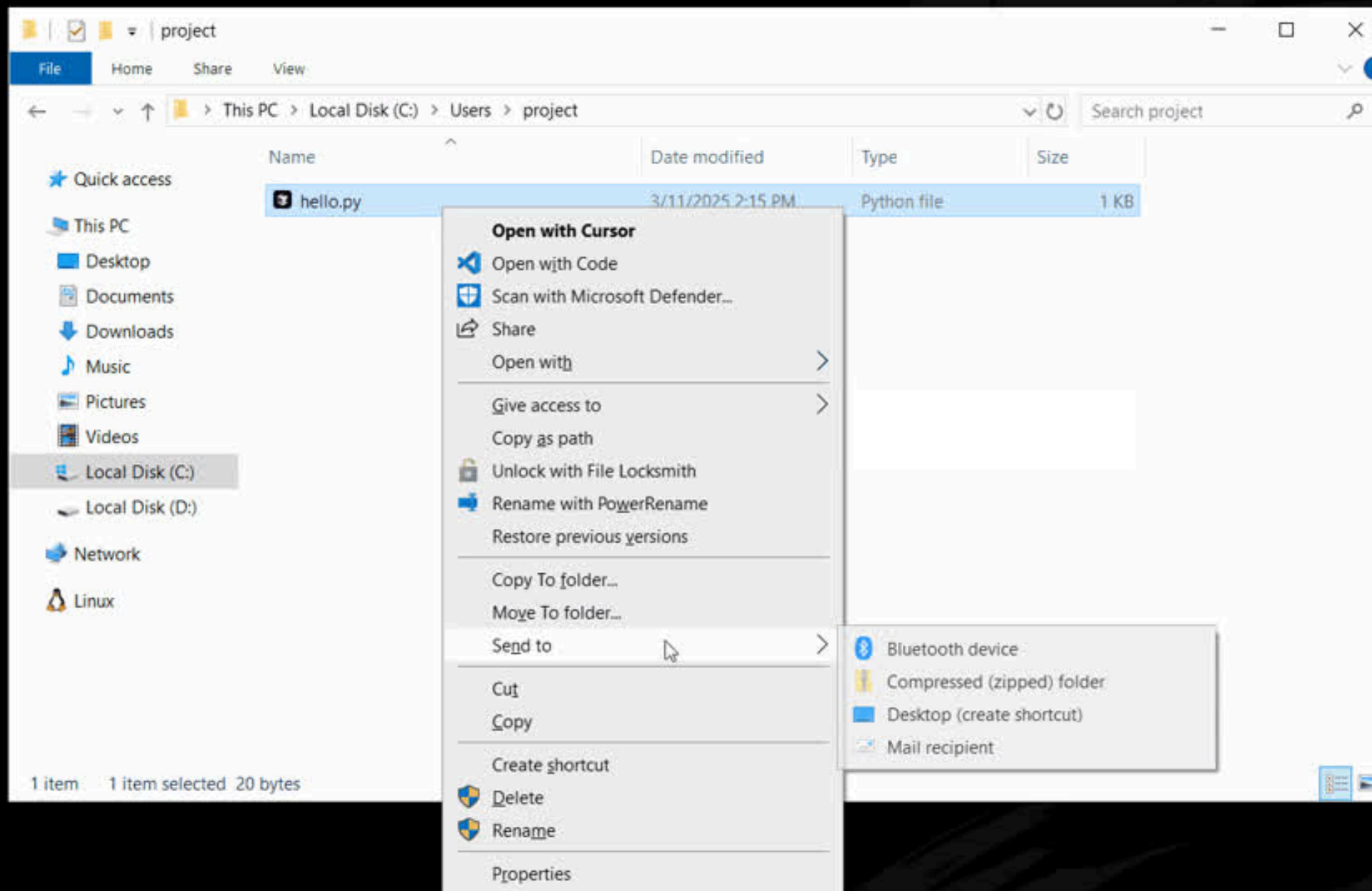
- ✓ You **need a fixed file location**.
- ✓ Accessing files **outside your project**.
- ✓ Running the script from any directory.

💡 Use Relative Path if:

- ✓ You want flexibility & portability.
- ✓ The script runs **within the project folder**.
- ✓ You prefer **shorter, cleaner file paths**.

Absolute Path

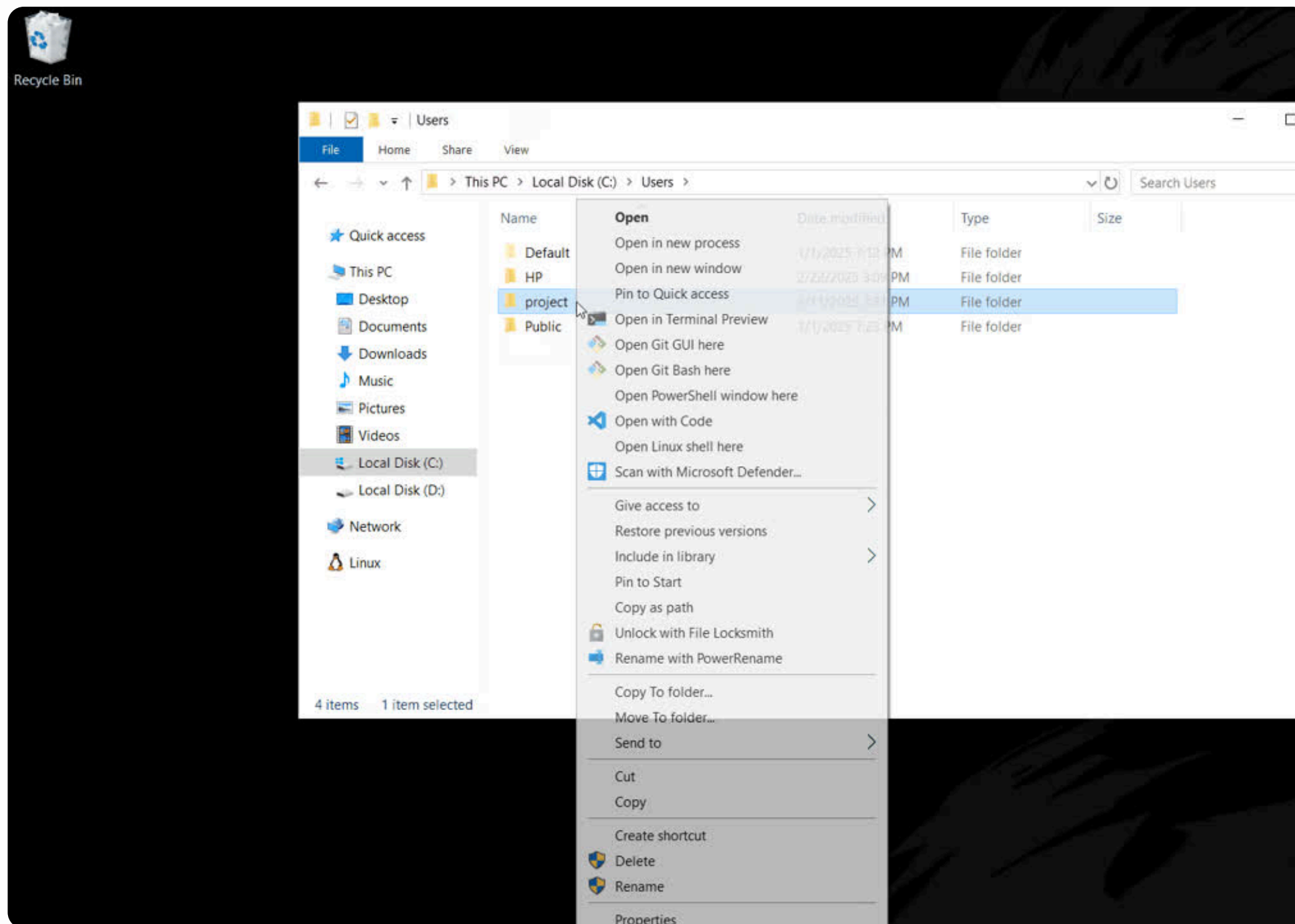
GamutX



- Go to **file location**
- Select **hello.py** file
- **Shift + Right Click** → Click "**Copy as Path**"
- Open **PowerShell**
- Type **python** and **paste the copied path**
- Hit **Enter** to run the script!

Relative Path

GamutX



Steps

- Go to the **project folder** location
- **Shift + Right Click** → Click "**Copy as Path**"
- Open **PowerShell**
- Type **cd** and **paste the copied folder path**
- Once in the **correct directory**,
 - Type **python**
 - **press Tab** to **auto-complete the script**
- Ensure command is **python .\hello.py**

- ◆ Let's Say our **file.py** is in **FC1 folder** which is located at **Desktop**

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\pralhad\Desktop\FC1> python file.py
Hello, World
PS C:\Users\pralhad\Desktop\FC1> python .\file.py
Hello, World
PS C:\Users\pralhad\Desktop\FC1> cd ..
PS C:\Users\pralhad\Desktop>
PS C:\Users\pralhad\Desktop>
PS C:\Users\pralhad\Desktop> python "C:\Users\pralhad\Desktop\FC1\file.py"
Hello, World
PS C:\Users\pralhad\Desktop> |
```

Using Python Keyword

Using Relative Path (dot)

Using Absolute Path

absolute path

Thank You

